13

Linear Differential Equations

13.1. DEFINITIONS

A **linear differential equation** is the differential equation in which the dependent variable and its derivatives occur only in the first degree and are not multiplied together. Thus, the general linear differential equation of the n^{th} order is of the form

$$\frac{d^n y}{dx^n} + P_1 \frac{d^{n-1}y}{dx^{n-1}} + P_2 \frac{d^{n-2}y}{dx^{n-2}} + \ldots\ldots + P_{n-1}\frac{dy}{dx} + P_n y = X,$$

where $P_1, P_2, \ldots\ldots, P_{n-1}, P_n$ and X are functions of x only.

A **linear differential equation with constant co-efficients** is of the form

$$\frac{d^n y}{dx^n} + a_1 \frac{d^{n-1}y}{dx^{n-1}} + a_2 \frac{d^{n-2}y}{dx^{n-2}} + \ldots\ldots + a_{n-1}\frac{dy}{dx} + a_n y = X \qquad \ldots(1)$$

where $a_1, a_2, \ldots\ldots, a_{n-1}, a_n$ are constants and X is either a constant or a function of x only.

13.2. THE OPERATOR D

The part $\frac{d}{dx}$ of the symbol $\frac{dy}{dx}$ may be regarded as an operator such that when it operates on y, the result is the derivative of y.

Similarly, $\frac{d^2}{dx^2}, \frac{d^3}{dx^3}, \ldots\ldots, \frac{d^n}{dx^n}$ may be regarded as operators.

For brevity, we write $\frac{d}{dx} \equiv D, \frac{d^2}{dx^2} \equiv D^2, \ldots\ldots, \frac{d^n}{dx^n} \equiv D^n$

Thus, the symbol D is a **differential operator** or simply an **operator.**

Written in symbolic form, equation (1) becomes

$$(D^n + a_1 D^{n-1} + a_2 D^{n-2} + \ldots\ldots + a_{n-1}D + a_n)y = X$$

or $\quad f(D)y = X$

where $\quad f(D) = D^n + a_1 D^{n-1} + a_2 D^{n-2} + \ldots\ldots + a_{n-1}D + a_n$

i.e., $f(D)$ is a polynomial in D.

The operator D can be treated as an algebraic quantity.

Thus $\quad D(u + v) = Du + Dv, \quad D(\lambda u) = \lambda\, Du$

$$D^p D^q\, u = D^{p+q}\, u, \qquad D^p D^q u = D^q D^p u$$

The polynomial $f(D)$ can be factorised by ordinary rules of algebra and the factors may be written in any order.

13.3. THEOREMS

Theorem 1. *If $y = y_1, y = y_2, \ldots, y = y_n$ are n linearly independent solutions of the differential equation*

$$(D^n + a_1 D^{n-1} + a_2 D^{n-2} + \ldots + a_n)y = 0 \qquad \ldots(i)$$

then $u = c_1 y_1 + c_2 y_2 + \ldots + c_n y_n$ is also its solution, where $c_1, c_2, \ldots, c_n$ are arbitrary constants.

Proof. Since $y = y_1, y = y_2, \ldots, y = y_n$ are solution of equation (*i*).

$$\therefore \left.\begin{array}{l} D^n y_1 + a_1 D^{n-1} y_1 + a_2 D^{n-2} y_1 + \ldots + a_n y_1 = 0 \\ D^n y_2 + a_1 D^{n-1} y_2 + a_2 D^{n-2} y_2 + \ldots + a_n y_2 = 0 \\ \ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots \\ \ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots \\ D^n y_n + a_1 D^{n-1} y_n + a_2 D^{n-2} y_n + \ldots + a_n y_n = 0 \end{array}\right\} \qquad \ldots(ii)$$

Now $D^n u + a_1 D^{n-1} u + a_2 D^{n-2} u + \ldots + a_n u$

$$\begin{aligned} &= D^n (c_1 y_1 + c_2 y_2 + \ldots + c_n y_n) \\ &\quad + a_1 D^{n-1}(c_1 y_1 + c_2 y_2 + \ldots + c_n y_n) + a_2 D^{n-2}(c_1 y_1 + c_2 y_2 + \ldots + c_n y_n) \\ &\quad + \quad \ldots \quad \ldots \quad \ldots \quad \ldots + a_n(c_1 y_1 + c_2 y_2 + \ldots + c_n y_n) \\ &= c_1(D^n y_1 + a_1 D^{n-1} y_1 + a_2 D^{n-2} y_1 + \ldots + a_n y_1) \\ &\quad + c_2(D^n y_2 + a_1 D^{n-1} y_2 + a_2 D^{n-2} y_n + \ldots + a_n y_2) \\ &\quad + \quad \ldots \quad \ldots + c_n(D^n y_n + a_1 D^{n-1} y_n + a_2 D^{n-2} y_n + \ldots + a_n y_n) \\ &= c_1(0) + c_2(0) + \ldots + c_n(0) \qquad \text{[From } (ii)] \\ &= 0 \end{aligned}$$

which shows that $u = c_1 y_1 + c_2 y_2 + \ldots\ldots + c_n y_n$ is also the solution of equation (*i*).

Since this solution contains n arbitrary constants, it is the general or complete solution of equation (*i*).

Theorem 2. *If $y = u$ is the complete solution of the equation $f(D)y = 0$ and $y = v$ is a particular solution (containing no arbitrary constants) of the equation $f(D)y = X$, then the complete solution of the equation $f(D)y = X$ is $y = u + v$.*

Proof. Since $y = u$ is the complete solution of the equation $f(D)y = 0$ ...(*i*)

$\therefore \quad f(D)u = 0$ ...(*ii*)

Also $y = v$ is a particular solution of the equation $f(D)y = X$ ...(*iii*)

$\therefore \quad f(D)v = X$ ...(*iv*)

Adding (*ii*) and (*iv*), we have $f(D)(u + v) = X$

Thus $y = u + v$ satisfies the equation (*iii*), hence it is the **complete solution (C.S.)** because it contains n arbitrary constants.

The part $y = u$ is called the **complementary function (C.F.)** and the part $y = v$ is called the **particular integral (P.I.)** of the equation (*iii*).

$\therefore$ The complete solution of equation (*iii*), is $\boxed{\mathbf{y = C.F. + P.I.}}$

Thus in order to solve the equation (iii), we first find the C.F. i.e., the C.S. of equation (i) and then the P.I. i.e., a particular solution of equation (iii).

13.4. AUXILIARY EQUATION (A.E.)

Consider the differential equation $(D^n + a_1 D^{n-1} + a_2 D^{n-2} + \ldots\ldots + a_n)y = 0$ ...(*i*)

Let $y = e^{mx}$ be a solution of (*i*), then $Dy = me^{mx}$, $D^2 y = m^2 e^{mx}$,......, $D^{n-2} y = m^{n-2} e^{mx}$

$$D^{n-1}y = m^{n-1}e^{mx},\ D^n y = m^n e^{mx}$$

Substituting the values of y, Dy, D^2y, ..., D^ny in (*i*), we get

$$(m^n + a_1 m^{n-1} + a_2 m^{n-2} + \ldots\ldots + a_n)\, e^{mx} = 0$$

or $$m^n + a_1 m^{n-1} + a_2 m^{n-2} + \ldots\ldots + a_n = 0, \text{ since } e^{mx} \neq 0 \qquad \ldots(ii)$$

Thus $y = e^{mx}$ will be a solution of equation (*i*) if m satisfies equation (*ii*).

Equation (*ii*) is called the auxiliary equation for the differential equation (*i*).

Replacing m by D in (*ii*), we get $D^n + a_1 D^{n-1} + a_2 D^{n-2} + \ldots\ldots + a_n = 0$ ...(*iii*)

Equation (*ii*) gives the same values of m *as* equation *(iii)* gives of D. In practice, we take equation (*iii*) as the auxiliary equation which is obtained by equating to zero the symbolic coefficient of y in equation (*i*).

Definition. The equation obtained by equating to zero the symbolic co-efficient of y is called the **auxiliary equation**, briefly written as **A.E.**

13.5. RULES FOR FINDING THE COMPLEMENTARY FUNCTION

Consider the equation $(D^n + a_1 D^{n-1} + a_2 D^{n-2} + \ldots\ldots + a_n)y = 0$ ...(*i*)

where all the a_i's are constant.

Its auxiliary equation is $D^n + a_1 D^{n-1} + a_2 D^{n-2} + \ldots\ldots + a_n = 0$ ...(*ii*)

Let $D = m_1, m_2, m_3, \ldots\ldots, m_n$ be the roots of the A.E. The solution of equation (*i*) depends upon the nature of roots of the A.E. The following cases arise :

Case I. *If all the roots of the A.E. are real and distinct,* then equation (*ii*) is equivalent to

$$(D - m_1)(D - m_2) \ldots\ldots (D - m_n) = 0 \qquad \ldots(iii)$$

Equation (*iii*) will be satisfied by the solutions of the equations

$$(D - m_1)y = 0,\ (D - m_2)y = 0, \ldots\ldots, (D - m_n)y = 0$$

Now, consider the equation $(D - m_1)y = 0$, *i.e.,* $\dfrac{dy}{dx} - m_1 y = 0$

It is a linear equation and I.F. $= e^{\int -m_1 dx} = e^{-m_1 x}$

$\therefore$ its solution is $y \,.\, e^{-m_1 x} = \int 0 \,.\, e^{-m_1 x}\, dx + c_1$ or $y = c_1 e^{m_1 x}$

Similarly, the solution of $(D - m_2)y = 0$ is $y = c_2 e^{m_2 x}$

....

the solution of $(D - m_n)y = 0$ is $y = c_n e^{m_n x}$

Hence the complete solution of equation (*i*) is

$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x} + \ldots\ldots + c_n e^{m_n x} \qquad \ldots(iv)$$

Case II. *If two roots of the A.E. are equal,* let $m_1 = m_2$.

The solution obtained in equation (*iv*) becomes

$$y = (c_1 + c_2)\, e^{m_1 x} + c_3 e^{m_3 x} + \ldots\ldots + c_n e^{m_n x}$$

$$= c e^{m_1 x} + c_3 e^{m_3 x} + \ldots\ldots + c_n e^{m_n x}$$

It contains $(n - 1)$ arbitrary constants and is, therefore, not the complete solution of equation (*i*).

The part of the complete solution corresponding to the repeated root is the complete solution of

$$(D - m_1)(D - m_1)y = 0$$

Putting $(D - m_1)y = v$, it becomes $(D - m_1)v = 0$ *i.e.*, $\frac{dv}{dx} - m_1 v = 0$

As in case I, its solution is $v = c_1 e^{m_1 x}$

$\therefore$ $$(D - m_1)y = c_1 e^{m_1 x} \quad \text{or} \quad \frac{dy}{dx} - m_1 y = c_1 e^{m_1 x}$$

which is a linear equation and I.F. $= e^{-m_1 x}$

$\therefore$ its solution is $y \,.\, e^{-m_1 x} = \int c_1 e^{m_1 x} . e^{-m_1 x}\, dx + c_2 = c_1 x + c_2$

or $$y = (c_1 x + c_2)\, e^{m_1 x}$$

Thus, the complete solution of equation (*i*) is

$$y = (c_1 x + c_2)\, e^{m_1 x} + c_3 e^{m_3 x} + \ldots\ldots + c_n e^{m_n x}$$

If, however, three roots of the A.E. are equal, say $m_1 = m_2 = m_3$, then proceeding as above, the solution becomes

$$y = (c_1 x^2 + c_2 x + c_3)\, e^{m_1 x} + c_4 e^{m_4 x} + \ldots\ldots + c_n e^{m_n x}$$

Case III. *If two roots of the A.E. are imaginary,* let

$$m_1 = \alpha + i\beta \quad \text{and} \quad m_2 = \alpha - i\beta$$

The solution obtained in equation (*iv*) becomes

$$\begin{aligned} y &= c_1 e^{(\alpha + i\beta)x} + c_2 e^{(\alpha - i\beta)x} + c_3 e^{m_3 x} + \ldots\ldots + c_n e^{m_n x} \\ &= e^{\alpha x}(c_1 e^{i\beta x} + c_2 e^{-i\beta x}) + c_3 e^{m_3 x} + \ldots\ldots + c_n e^{m_n x} \\ &= e^{\alpha x}[c_1(\cos \beta x + i \sin \beta x) + c_2 (\cos \beta x - i \sin \beta x)] \\ &\quad + c_3 e^{m_3 x} + \ldots\ldots + c_n e^{m_n x} \quad [\because \text{ by Euler's Theorem, } e^{i\theta} = \cos\theta + i \sin\theta] \\ &= e^{\alpha x}[(c_1 + c_2) \cos \beta x + i\,(c_1 - c_2) \sin \beta x] + c_3 e^{m_3 x} + \ldots\ldots + c_n e^{m_n x} \\ &= e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x) + c_3 e^{m_3 x} + \ldots\ldots + c_n e^{m_n x} \end{aligned}$$

[taking $c_1 + c_2 = C_1$, $i(c_1 - c_2) = C_2$]

Case IV. *If two pairs of imaginary roots be equal,* let

$$m_1 = m_2 = \alpha + i\beta \quad \text{and} \quad m_3 = m_4 = \alpha - i\beta$$

Then by case II, the complete solution is

$$y = e^{\alpha x}\, [(c_1 x + c_2) \cos \beta x + (c_3 x + c_4) \sin \beta x] + c_5 e^{m_5 x} + \ldots\ldots + c_n e^{m_n x}.$$

ILLUSTRATIVE EXAMPLES

Example 1. *Solve* $\frac{d^3 y}{dx^3} - 7\frac{dy}{dx} - 6y = 0.$

Sol. Given equation in symbolic form is $(D^3 - 7D - 6)y = 0$

Its A.E. is $m^3 - 7m - 6 = 0$ or $(m + 1)(m + 2)(m - 3) = 0$

whence $m = -1, -2, 3$

Hence the C.S. is $y = c_1 e^{-x} + c_2 e^{-2x} + c_3 e^{3x}$.

Example 2. *Solve* $(D^3 - 4D^2 + 4D)y = 0$.

Sol. The A.E. is $m^3 - 4m^2 + 4m = 0$ or $m(m^2 - 4m + 4) = 0$

or $m(m - 2)^2 = 0$

whence $m = 0, 2, 2$

Hence, the C.S is $y = c_1 e^{0x} + (c_2 x + c_3)e^{2x}$ or $y = c_1 + (c_2 x + c_3)e^{2x}$.

Example 3. *Solve* $\dfrac{d^4 y}{dx^4} + 13\dfrac{d^2 y}{dx^2} + 36y = 0.$

Sol. Given equation in symbolic form is $(D^4 + 13D^2 + 36)y = 0$

Its A.E. is $m^4 + 13m^2 + 36 = 0$

or $(m^2 + 4)(m^2 + 9) = 0$ or $m = \pm 2i, \pm 3i$

Hence the C.S. is $y = e^{0x}(c_1 \cos 2x + c_2 \sin 2x) + e^{0x}(c_3 \cos 3x + c_4 \sin 3x)$

or $y = c_1 \cos 2x + c_2 \sin 2x + c_3 \cos 3x + c_4 \sin 3x$.

Example 4. *Solve* $\dfrac{d^4 x}{dt^4} + 4x = 0.$

Sol. Given equation in symbolic form is $(D^4 + 4)x = 0$, where $D \equiv \dfrac{d}{dt}$

Its A.E. is $m^4 + 4 = 0$ or $(m^4 + 4m^2 + 4) - 4m^2 = 0$

or $(m^2 + 2)^2 - (2m)^2 = 0$ or $(m^2 + 2m + 2)(m^2 - 2m + 2) = 0$

whence $m = \dfrac{-2 \pm \sqrt{-4}}{2}$ and $\dfrac{2 \pm \sqrt{-4}}{2}$ *i.e.,* $m = -1 \pm i$ and $1 \pm i$

Hence the C.S. is $x = e^{-t}(c_1 \cos t + c_2 \sin t) + e^{t}(c_3 \cos t + c_4 \sin t)$.

TEST YOUR KNOWLEDGE

Solve the following differential equations :

1. $\dfrac{d^2 y}{dx^2} - 3\dfrac{dy}{dx} - 4y = 0$.

2. $\dfrac{d^2 y}{dx^2} + (a + b)\dfrac{dy}{dx} + aby = 0$.

3. $\dfrac{d^2 y}{dx^2} - 4\dfrac{dy}{dx} + y = 0$.

4. $\dfrac{d^2 x}{dt^2} + 6\dfrac{dx}{dt} + 9x = 0$.

5. $\dfrac{d^3 y}{dx^3} - 3\dfrac{d^2 y}{dx^2} + 3\dfrac{dy}{dx} - y = 0$.

6. $\dfrac{d^3 y}{dx^3} + 6\dfrac{d^2 y}{dx^2} + 11\dfrac{dy}{dx} + 6y = 0$.

7. $\dfrac{d^4 y}{dx^4} - 5\dfrac{d^2 y}{dx^2} + 4y = 0$.

8. $\dfrac{d^4 y}{dx^4} + 8\dfrac{d^2 y}{dx^2} + 16y = 0$.

9. $(D^2 + 1)^3 (D^2 + D + 1)^2 y = 0$.

10. $\dfrac{d^2 x}{dt^2} - 3\dfrac{dx}{dt} + 2x = 0$, given that when $t = 0$, $x = 0$ and $\dfrac{dx}{dt} = 0$.

11. $\dfrac{d^2y}{dx^2} + 4\dfrac{dy}{dx} + 29y = 0$, given that when $x = 0$, $y = 0$ and $\dfrac{dy}{dx} = 15$.

12. If $\dfrac{d^4x}{dt^4} = m^4x$, show that $x = c_1 \cos mt + c_2 \sin mt + c_3 \cosh mt + c_4 \sinh mt$.

Answers

1. $y = c_1 e^{4x} + c_2 e^{-x}$ **2.** $y = c_1 e^{-ax} + c_2 e^{-bx}$ **3.** $y = c_1 e^{(2+\sqrt{3})x} + c_2 e^{(2-\sqrt{3})x}$

4. $x = (c_1 + c_2 t)\, e^{-3t}$ **5.** $y = (c_1 + c_2 x + c_3 x^2)\, e^x$ **6.** $y = c_1 e^{-x} + c_2 e^{-2x} + c_3 e^{-3x}$

7. $y = c_1 e^x + c_2 e^{-x} + c_3 e^{2x} + c_4 e^{-2x}$ **8.** $y = (c_1 + c_2 x) \cos 2x + (c_3 + c_4 x) \sin 2x$

9. $y = (c_1 + c_2 x + c_3 x^2) \cos x + (c_4 + c_5 x + c_6 x^2) \sin x + e^{-\frac{1}{2}x} \left[(c_7 + c_8 x) \cos \dfrac{\sqrt{3}}{2} x + (c_9 + c_{10} x) \sin \dfrac{\sqrt{3}}{2} x \right]$

10. $x = 0$ **11.** $y = 3e^{-2x} \sin 5x$.

13.6. THE INVERSE OPERATOR $\dfrac{1}{f(D)}$

Definition. $\dfrac{1}{f(\mathrm{D})}\mathrm{X}$ is that function of x, *free from arbitrary constants,* which when operated upon by $f(\mathrm{D})$ gives X.

Thus $f(\mathrm{D}) \left\{ \dfrac{1}{f(\mathrm{D})} \mathrm{X} \right\} = \mathrm{X}$

$\therefore$ $f(\mathrm{D})$ and $\dfrac{1}{f(\mathrm{D})}$ are inverse operators.

Theorem 1. $\dfrac{1}{f(D)}X$ *is the particular integral of* $f(D)y = X$.

Proof. The given equation is $f(\mathrm{D})y = \mathrm{X}$ ...(1)

Putting $y = \dfrac{1}{f(\mathrm{D})}\mathrm{X}$ in (1), we have $f(\mathrm{D}) \left\{ \dfrac{1}{f(\mathrm{D})} \mathrm{X} \right\} = \mathrm{X}$ or $\mathrm{X} = \mathrm{X}$, which is true.

$\therefore$ $y = \dfrac{1}{f(\mathrm{D})}\mathrm{X}$ is a solution of (1).

Since it contains no arbitrary constants, it is the particular integral of $f(\mathrm{D})\, y = \mathrm{X}$.

Theorem 2. *Prove that,* $\dfrac{1}{D}X = \int X\, dx$.

Proof. Let $\dfrac{1}{\mathrm{D}}\mathrm{X} = y$

Operating both sides by D, we have $\mathrm{D}\left(\dfrac{1}{\mathrm{D}}\mathrm{X}\right) = \mathrm{D}y$ or $\mathrm{X} = \dfrac{dy}{dx}$

Integrating both sides w.r.t. x,

$$y = \int \mathrm{X}\, dx,$$

no arbitrary constant being added since $y = \frac{1}{D}X$ contains no arbitrary constant.

$$\therefore \qquad \frac{1}{D}X = \int X\, dx.$$

Theorem 3. *Prove that,* $\frac{1}{D-a}X = e^{ax}\int Xe^{-ax}\, dx$.

Proof. Let $\frac{1}{D-a}X = y$

Operating on both sides by $(D - a)$, $(D-a)\left(\frac{1}{D-a}X\right) = (D-a)y$

or $$X = \frac{dy}{dx} - ay \quad i.e., \quad \frac{dy}{dx} - ay = X$$

which is a linear equation and I.F. $= e^{\int -a\,dx} = e^{-ax}$

$\therefore$ Its solution is $ye^{-ax} = \int X e^{-ax}\, dx$, no constant being added or $y = e^{ax}\int X e^{-ax}\, dx$

Hence $$\frac{1}{D-a}X = e^{ax}\int e^{-ax}\, X\, dx.$$

13.7. RULES FOR FINDING THE PARTICULAR INTEGRAL

Consider the differential equation, $(D^n + a_1D^{n-1} + a_2D^{n-2} + \ldots\ldots + a_{n-1}D + a_n)y = X$

It can be written as $f(D)y = X$

$$\therefore \qquad \text{P.I.} = \frac{1}{f(D)}X$$

Case I. When $X = e^{ax}$

Since
$$D\, e^{ax} = a\, e^{ax}$$
$$D^2\, e^{ax} = a^2\, e^{ax}$$
$$\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$
$$\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$
$$D^{n-1}\, e^{ax} = a^{n-1}\, e^{ax}$$
$$D^n e^{ax} = a^n\, e^{ax}$$

$$\therefore \quad (D^n + a_1\, D^{n-1} + a_2\, D^{n-2} + \ldots\ldots + a_{n-1}D + a_n)\, e^{ax}$$
$$= (a^n + a_1\, a^{n-1} + a_2\, a^{n-2} + \ldots\ldots + a_{n-1}\, a + a_n)\, e^{ax}$$

or $$f(D)\, e^{ax} = f(a)\, e^{ax}$$

Operating on both sides by $\frac{1}{f(D)}$.

$$\frac{1}{f(D)}(f(D)e^{ax}) = \frac{1}{f(D)}(f(a)e^{ax}) \quad \text{or} \quad e^{ax} = f(a)\,\frac{1}{f(D)}e^{ax}$$

Dividing both sides by $f(a)$, $\frac{1}{f(a)}e^{ax} = \frac{1}{f(D)}e^{ax}$, provided $f(a) \neq 0$

Hence $$\mathbf{\frac{1}{f(D)}e^{ax} = \frac{1}{f(a)}e^{ax},} \qquad \textbf{provided } \mathbf{f(a) \neq 0.}$$

Case of failure. If $f(a) = 0$, the above method fails.

Since $f(a) = 0$, $D = a$ is a root of A.E. $f(D) = 0$

$\therefore$ $D - a$ is a factor of $f(D)$.

Let $$f(D) = (D - a)\,\phi(D), \text{ where } \phi(a) \neq 0 \qquad ...(i)$$

Then $$\frac{1}{f(D)}e^{ax} = \frac{1}{(D-a)\,\phi(D)}e^{ax} = \frac{1}{D-a}\cdot\frac{1}{\phi(D)}e^{ax} = \frac{1}{D-a}\cdot\frac{1}{\phi(a)}e^{ax}$$

$$= \frac{1}{\phi(a)}\cdot\frac{1}{D-a}e^{ax} = \frac{1}{\phi(a)}e^{ax}\int e^{ax}\,.\,e^{-ax}dx \qquad \text{[by Theorem 3]}$$

$$= \frac{1}{\phi(a)}e^{ax}\int 1\,dx = x\,.\,\frac{1}{\phi(a)}e^{ax} \qquad ...(ii)$$

Differentiating both sides of (*i*) w.r.t. D, we have $f'(D) = (D - a)\,\phi'(D) + \phi(D)$

$\Rightarrow$ $$f'(a) = \phi(a)$$

$\therefore$ From (*ii*), we have $\mathbf{\dfrac{1}{f(D)}e^{ax} = x\,.\,\dfrac{1}{f'(a)}e^{ax}}$, **provided $\mathbf{f'(a) \neq 0}$**

If $f'(a) = 0$, then $\dfrac{1}{f(D)}e^{ax} = x^2\,.\,\dfrac{1}{f''(a)}e^{ax}$, provided $f''(a) \neq 0$

and so on.

ILLUSTRATIVE EXAMPLES

Example 1. *Find the P.I. of $(4D^2 + 4D - 3)\,y = e^{2x}$.*

Sol. $$\text{P.I.} = \frac{1}{4D^2 + 4D - 3}e^{2x} = \frac{1}{4(2)^2 + 4(2) - 3}e^{2x} \qquad \text{(replacing D by 2)}$$

$$= \frac{1}{21}e^{2x}.$$

Example 2. *Find the P.I. of $(D^2 + 3D + 2)y = 5$.*

Sol. $$\text{P.I.} = \frac{1}{D^2 + 3D + 2}(5e^{0x}) \qquad [\because\ e^{0x} = 1]$$

$$= 5\,.\,\frac{1}{0 + 0 + 2}e^{0x} = \frac{5}{2}. \qquad \text{(replacing D by 0)}$$

Example 3. *Find the P.I. of $(D^3 - 3D^2 + 4)y = e^{2x}$.*

Sol. $$\text{P.I.} = \frac{1}{D^3 - 3D^2 + 4}e^{2x}.$$

Hence the denominator vanishes when D is replaced by 2. It is a case of failure.

We multiply the numerator by x and differentiate the denominator w.r.t. D.

$\therefore$ $$\text{P.I.} = x\,.\,\frac{1}{3D^2 - 6D}e^{2x}$$

It is again a case of failure. We multiply the numerator by x and differentiate the denominator w.r.t. D.

$$\therefore \quad \text{P.I.} = x^2 \cdot \frac{1}{6D-6} e^{2x} = x^2 \cdot \frac{1}{6(2)-6} e^{2x} = \frac{x^2}{6} e^{2x}.$$

Case II. When X = sin (ax + b) or cos (ax + b)

$$D \sin(ax+b) = a \cos(ax+b)$$

$$\mathbf{D^2 \sin(ax+b) = (-a^2) \sin(ax+b)}$$

$$D^3 \sin(ax+b) = -a^3 \cos(ax+b)$$

$$D^4 \sin(ax+b) = a^4 \sin(ax+b)$$

or $$\mathbf{(D^2)^2 \sin(ax+b) = (-a^2)^2 \sin(ax+b)}$$

....

....

In general, $(D^2)^n \sin(ax+b) = (-a^2)^n \sin(ax+b)$

$\therefore \quad f(D^2) \sin(ax+b) = f(-a^2) \sin(ax+b)$

Operating on both sides by $\frac{1}{f(D^2)}$,

$$\frac{1}{f(D^2)} (f(D^2) \sin(ax+b)) = \frac{1}{f(D^2)} (f(-a^2) \sin(ax+b))$$

or $$\sin(ax+b) = f(-a^2) \frac{1}{f(D^2)} \sin(ax+b).$$

Dividing both sides by $f(-a^2)$,

$$\frac{1}{f(-a^2)} \sin(ax+b) = \frac{1}{f(D^2)} \sin(ax+b), \qquad \text{provided } f(-a^2) \neq 0.$$

Hence $$\mathbf{\frac{1}{f(D^2)} \sin(ax+b) = \frac{1}{f(-a^2)} \sin(ax+b), \qquad provided\ f(-a^2) \neq 0}$$

Similarly, $$\mathbf{\frac{1}{f(D^2)} \cos(ax+b) = \frac{1}{f(-a^2)} \cos(ax+b), \qquad provided\ f(-a^2) \neq 0}$$

Case of failure. If $f(-a^2) = 0$, the above method fails.

Since $\cos(ax+b) + i \sin(ax+b) = e^{i(ax+b)}$ | Euler's Theorem

$$\therefore \quad \frac{1}{f(D^2)} [\cos(ax+b) + i \sin(ax+b)] = \frac{1}{f(D^2)} e^{i(ax+b)}$$

[If we replace D by ia, $f(D^2) = f(-a^2) = 0$, so that it is a case of failure]

$$= x \cdot \frac{1}{f'(D^2)} e^{i(ax+b)} = x \cdot \frac{1}{f'(D^2)} [\cos(ax+b) + i \sin(ax+b)]$$

Equating real parts

$$\mathbf{\frac{1}{f(D^2)} \cos(ax+b) = x \cdot \frac{1}{f'(D^2)} \cos(ax+b), \qquad provided\ f'(-a^2) \neq 0}$$

Equating imaginary parts

$$\mathbf{\frac{1}{f(D^2)} \sin(ax+b) = x \cdot \frac{1}{f'(D^2)} \sin(ax+b), \qquad provided\ f'(-a^2) \neq 0}$$

If $f'(-a^2) = 0$, then

$$\frac{1}{f(D^2)} \sin(ax+b) = x^2 \cdot \frac{1}{f''(D^2)} \sin(ax+b), \qquad \text{provided } f''(-a^2) \neq 0$$

$$\frac{1}{f(D^2)} \cos(ax+b) = x^2 \cdot \frac{1}{f''(D^2)} \cos(ax+b), \qquad \text{provided } f''(-a^2) \neq 0$$

and so on.

ILLUSTRATIVE EXAMPLES

Example 1. *Find the P.I. of $(D^3 + 1)y = \sin(2x + 3)$.*

Sol.

$$\text{P.I.} = \frac{1}{D^3+1} \sin(2x+3) = \frac{1}{D(-2^2)+1} \sin(2x+3) \qquad [\text{Putting } D^2 = -2^2]$$

$$= \frac{1}{1-4D} \sin(2x+3)$$

Multiplying and dividing by $(1 + 4D)$

$$= \frac{1+4D}{(1-4D)(1+4D)} \sin(2x+3) = \frac{1+4D}{1-16D^2} \sin(2x+3)$$

$$= \frac{1+4D}{1-16(-2^2)} \sin(2x+3) \qquad [\text{Putting } D^2 = -2^2]$$

$$= \frac{1}{65} [\sin(2x+3) + 4D \sin(2x+3)]$$

$$= \frac{1}{65} [\sin(2x+3) + 8\cos(2x+3)] \qquad \left[\because D = \frac{d}{dx}\right]$$

Example 2. *Find the P.I. of $(D^2 + 4)y = \cos 2x$.*

Sol.

$$\text{P.I.} = \frac{1}{D^2+4} \cos 2x$$

Here the denominator vanishes when D^2 is replaced by $-2^2 = -4$. It is a case of failure. We multiply the numerator by x and differentiate the denominator w.r.t. D.

$$\therefore \qquad \text{P.I.} = x \cdot \frac{1}{2D} \cos 2x = \frac{x}{2} \int \cos 2x \, dx \qquad \left[\because \frac{1}{D} f(x) = \int f(x) dx\right]$$

$$= \frac{x}{4} \sin 2x.$$

Case III. When $X = x^m$, m being a positive integer.

Here $\qquad \text{P.I.} = \dfrac{1}{f(D)} x^m$

Take out the lowest degree term from $f(D)$ to make the first term unity (so that Binomial Theorem for a negative index is applicable).

The remaining factor will be of the form $1 + \phi(D)$ or $1 - \phi(D)$

Take this factor in the numerator. It takes the form $[1 + \phi(D)]^{-1}$ or $[1 - \phi(D)]^{-1}$

Expand it in ascending powers of D as far as the term containing D^m, since $D^{m+1}(x^m) = 0$, $D^{m+2}(x^m) = 0$ and so on.

Operate on x^m term by term.

Example. *Find the P.I. of* $(D^2 + 5D + 4)y = x^2 + 7x + 9$.

Sol.
$$\text{P.I.} = \frac{1}{D^2+5D+4}(x^2+7x+9) = \frac{1}{4\left(1+\frac{5D}{4}+\frac{D^2}{4}\right)}(x^2+7x+9)$$

$$= \frac{1}{4}\left[1+\left(\frac{5D}{4}+\frac{D^2}{4}\right)\right]^{-1}(x^2+7x+9)$$

$$= \frac{1}{4}\left[1-\left(\frac{5D}{4}+\frac{D^2}{4}\right)+\left(\frac{5D}{4}+\frac{D^2}{4}\right)^2-\ldots\ldots\right](x^2+7x+9)$$

$$= \frac{1}{4}\left(1-\frac{5D}{4}-\frac{D^2}{4}+\frac{25D^2}{16}\ldots\ldots\right)(x^2+7x+9)$$

$$= \frac{1}{4}\left(1-\frac{5D}{4}+\frac{21D^2}{16}\ldots\ldots\right)(x^2+7x+9)$$

$$= \frac{1}{4}\left[(x^2+7x+9)-\frac{5}{4}D(x^2+7x+9)+\frac{21}{16}D^2(x^2+7x+9)\right]$$

$$= \frac{1}{4}\left[(x^2+7x+9)-\frac{5}{4}(2x+7)+\frac{21}{16}(2)\right] = \frac{1}{4}\left(x^2+\frac{9}{2}x+\frac{23}{8}\right).$$

Case IV. When $X = e^{ax} V$, where V is a function of x.

Let u be a function of x, then by successive differentiation, we have

$$D(e^{ax}u) = e^{ax}Du + a\,e^{ax}u = e^{ax}(D+a)u$$

$$D^2(e^{ax}u) = D\,[e^{ax}(D+a)u] = e^{ax}(D^2+aD)u + ae^{ax}(D+a)u$$

$$= e^{ax}(D^2+2aD+a^2)u = e^{ax}(D+a)^2u$$

Similarly, $\quad D^3(e^{ax}u) = e^{ax}(D+a)^3u$

In general, $\quad D^n(e^{ax}u) = e^{ax}(D+a)^nu$

$\therefore \quad f(D)(e^{ax}u) = e^{ax}f(D+a)u$

Operating on both sides by $\dfrac{1}{f(D)}$,

$$\frac{1}{f(D)}[f(D)(e^{ax}u)] = \frac{1}{f(D)}[e^{ax}f(D+a)u]$$

$$\Rightarrow \quad e^{ax}u = \frac{1}{f(D)}[e^{ax}f(D+a)u] \qquad \ldots(i)$$

Now let $\quad f(D+a)u = V, \quad i.e., \quad u = \dfrac{1}{f(D+a)}V$

$\therefore$ From (*i*), we have

or $$\frac{1}{\mathbf{f(D)}}(\mathbf{e^{ax}\,V}) = \mathbf{e^{ax}}\,\frac{1}{\mathbf{f(D+a)}}\mathbf{V}.$$

Thus e^{ax} which is on the right of $\dfrac{1}{f(D)}$ may be taken out to the left provided D is replaced by $D + a$.

Example. *Find the P.I. of* $(D^2 - 4D + 3)y = e^x \cos 2x$.

Sol.
$$\text{P.I.} = \frac{1}{D^2 - 4D + 3}\, e^x \cos 2x = e^x \frac{1}{(D+1)^2 - 4(D+1) + 3} \cos 2x$$

$$= e^x \frac{1}{D^2 - 2D} \cos 2x = e^x \frac{1}{-2^2 - 2D} \cos 2x \qquad [\text{Putting } D^2 = -2^2]$$

$$= -\frac{1}{2} e^x \frac{1}{2 + D} \cos 2x$$

$$= -\frac{1}{2} e^x \frac{2 - D}{4 - D^2} \cos 2x = -\frac{1}{2} e^x \frac{2 - D}{4 - (-2^2)} \cos 2x$$

$$= -\frac{1}{16} e^x (2 \cos 2x - D \cos 2x) = -\frac{1}{16} e^x (2 \cos 2x + 2 \sin 2x)$$

$$= -\frac{1}{8} e^x (\cos 2x + \sin 2x).$$

Case V. When X is any other function of x.

Resolve $f(D)$ into linear factors.

Let $\quad f(D) = (D - m_1)(D - m_2) \ldots\ldots (D - m_n)\, X$

Then
$$\text{P.I.} = \frac{1}{f(D)} X = \frac{1}{(D - m_1)(D - m_2) \ldots\ldots (D - m_n)} X$$

$$= \left(\frac{A_1}{D - m_1} + \frac{A_2}{D - m_2} + \ldots\ldots + \frac{A_n}{D - m_n}\right) X \qquad \text{(Partial Fractions)}$$

$$= A_1 \frac{1}{D - m_1} X + A_2 \frac{1}{D - m_2} X + \ldots\ldots + A_n \frac{1}{D - m_n} X$$

$$= A_1 e^{m_1 x} \int X e^{-m_1 x}\, dx + A_2 e^{m_2 x} \int X e^{-m_2 x} dx + \ldots\ldots + A_n e^{m_n x} \int X e^{-m_n x} dx$$

ILLUSTRATIVE EXAMPLES

Example 1. *Solve* $(D^3 - 6D^2 + 11D - 6)y = e^{-2x} + e^{-3x}$.

Sol. A.E. is $= m^3 - 6m^2 + 11m - 6 = 0$

whence $\quad m = 1, 2, 3$

$\therefore \quad \text{C.F.} = c_1 e^x + c_2 e^{2x} + c_3 e^{3x}$

$$\text{P.I.} = \frac{1}{D^3 - 6D^2 + 11D - 6} (e^{-2x} + e^{-3x})$$

$$= \frac{1}{D^3 - 6D^2 + 11D - 6} e^{-2x} + \frac{1}{D^3 - 6D^2 + 11D - 6} e^{-3x}$$

$$= \frac{1}{(-2)^3 - 6(-2)^2 + 11(-2) - 6} e^{-2x} + \frac{1}{(-3)^3 - 6(-3)^2 + 11(-3) - 6} e^{-3x}$$

$$= -\frac{1}{60} e^{-2x} - \frac{1}{120} e^{-3x} = -\frac{1}{120} (2e^{-2x} + e^{-3x})$$

Hence the C.S. is $y = \text{C.F.} + \text{P.I.} = c_1e^x + c_2e^{2x} + c_3e^{3x} - \frac{1}{120} (2e^{-2x} + e^{-3x})$.

Example 2. *Solve* $(D - 2)^2y = 8(e^{2x} + \sin 2x + x^2)$.

Sol. A.E. is $(m - 2)^2 = 0$ whence $m = 2, 2$

$$\therefore \qquad \text{C.F.} = (c_1 + c_2x)e^{2x}$$

$$\text{P.I.} = \frac{1}{(D-2)^2} [8(e^{2x} + \sin 2x + x^2)]$$

$$= 8\left[\frac{1}{(D-2)^2} e^{2x} + \frac{1}{(D-2)^2} \sin 2x + \frac{1}{(D-2)^2} x^2\right]$$

Now $\frac{1}{(D-2)^2} e^{2x} = x \cdot \frac{1}{2(D-2)} e^{2x} = x^2 \cdot \frac{1}{2} e^{2x}$ | Case of failure

$$\frac{1}{(D-2)^2} \sin 2x = \frac{1}{D^2 - 4D + 4} \sin 2x = \frac{1}{-2^2 - 4D + 4} \sin 2x \qquad [\text{Putting } D^2 = -2^2]$$

$$= -\frac{1}{4D} \sin 2x = -\frac{1}{4}\int \sin 2x \, dx = -\frac{1}{4}\left(-\frac{\cos 2x}{2}\right) = \frac{1}{8} \cos 2x$$

$$\frac{1}{(D-2)^2} x^2 = \frac{1}{(2-D)^2} x^2 = \frac{1}{4\left(1-\frac{D}{2}\right)^2} x^2 = \frac{1}{4}\left(1-\frac{D}{2}\right)^{-2} x^2$$

$$= \frac{1}{4}\left[1 - 2\left(-\frac{D}{2}\right) + \frac{(-2)(-3)}{2}\left(\frac{D}{2}\right)^2 \ldots\ldots\right] x^2$$

$$= \frac{1}{4}\left[1 + D + \frac{3}{4}D^2 + \ldots\ldots\right] x^2 = \frac{1}{4}\left(x^2 + 2x + \frac{3}{2}\right)$$

$$\therefore \qquad \text{P.I.} = 8\left[\frac{x^2}{2}e^{2x} + \frac{1}{8}\cos 2x + \frac{1}{4}\left(x^2 + 2x + \frac{3}{2}\right)\right]$$

$$= 4x^2 e^{2x} + \cos 2x + 2x^2 + 4x + 3$$

Hence the C.S. is $y = (c_1 + c_2x) e^{2x} + 4x^2 e^{2x} + \cos 2x + 2x^2 + 4x + 3$.

Example 3. *Solve* $(D + 2)(D - 1)^2 y = e^{-2x} + 2 \sinh x$.

Sol. A.E. is $(m + 2)(m - 1)^2 = 0$ so that $m = -2, 1, 1$

$$\therefore \qquad \text{C.F.} = c_1 e^{-2x} + (c_2 + c_3 x)\, e^x$$

$$\text{P.I.} = \frac{1}{(D+2)(D-1)^2}\,(e^{-2x} + 2\sinh x)$$

$$= \frac{1}{(D+2)(D-1)^2}\,(e^{-2x} + e^x - e^{-x}) \qquad \left[\because \ \sinh x = \frac{e^x - e^{-x}}{2}\right]$$

Now $\dfrac{1}{(D+2)(D-1)^2}\, e^{-2x} = \dfrac{1}{D+2}\left[\dfrac{1}{(D-1)^2}\, e^{-2x}\right] = \dfrac{1}{D+2}\left[\dfrac{1}{(-2-1)^2}\, e^{-2x}\right]$

$$= \frac{1}{9}\cdot\frac{1}{D+2}\, e^{-2x} \qquad |\ \text{Case of failure}$$

$$= \frac{x}{9} e^{-2x}$$

$$\frac{1}{(D+2)(D-1)^2}\, e^x = \frac{1}{(D-1)^2}\left[\frac{1}{D+2}\, e^x\right] = \frac{1}{(D-1)^2}\left[\frac{1}{1+2}\, e^x\right]$$

$$= \frac{1}{3}\cdot\frac{1}{(D-1)^2}\, e^x \qquad |\ \text{Case of failure}$$

$$= \frac{1}{3}\,.\, x\,\frac{1}{2(D-1)}\, e^x \qquad |\ \text{Case of failure}$$

$$= \frac{1}{3}\,.\, x^2\,.\,\frac{1}{2}\, e^x = \frac{1}{6}\, x^2 e^x$$

$$\frac{1}{(D+2)(D-1)^2}\, e^{-x} = \frac{1}{(-1+2)(-1-1)^2}\, e^{-x} = \frac{1}{4}\, e^{-x}$$

$$\therefore \qquad \text{P.I.} = \frac{x}{9}\, e^{-2x} + \frac{x^2}{6}\, e^x + \frac{1}{4}\, e^{-x}$$

Hence the C.S. is $\qquad y = c_1 e^{-2x} + (c_2 + c_3 x) e^x + \dfrac{x}{9}\, e^{-2x} + \dfrac{x^2}{6}\, e^x + \dfrac{1}{4}\, e^{-x}.$

Example 4. *Solve* $\dfrac{d^2 y}{dx^2} - 4y = x \sinh x.$

Sol. Given equation in symbolic form is $(D^2 - 4)y = x \sinh x$

A.E. is $\qquad m^2 - 4 = 0 \quad$ so that $\quad m = \pm 2$

$\therefore \quad \text{C.F.} = c_1 e^{2x} + c_2 e^{-2x}$

$$\text{P.I.} = \frac{1}{D^2 - 4}\, x \sinh x = \frac{1}{D^2 - 4}\, x\left(\frac{e^x - e^{-x}}{2}\right)$$

$$= \frac{1}{2}\left[\frac{1}{D^2 - 4}\, e^x\,.\,x - \frac{1}{D^2 - 4}\, e^{-x}\,.\,x\right]$$

$$= \frac{1}{2}\left[e^x\,\frac{1}{(D+1)^2 - 4}\, x - e^{-x}\,\frac{1}{(D-1)^2 - 4}\, x\right]$$

$$= \frac{1}{2}\left[e^{x}\frac{1}{D^2+2D-3}x - e^{-x}\frac{1}{D^2-2D-3}x\right]$$

$$= \frac{1}{2}\left[e^{x}\frac{1}{-3\left(1-\frac{2D}{3}-\frac{D^2}{3}\right)}x - e^{-x}\frac{1}{-3\left(1+\frac{2D}{3}-\frac{D^2}{3}\right)}x\right]$$

$$= -\frac{1}{6}\left[e^{x}\left\{1-\left(\frac{2D}{3}+\frac{D^2}{3}\right)\right\}^{-1}x - e^{-x}\left\{1+\left(\frac{2D}{3}-\frac{D^2}{3}\right)\right\}^{-1}x\right]$$

$$= -\frac{1}{6}\left[e^{x}\left(1+\frac{2D}{3}\ldots\ldots\right)x - e^{-x}\left(1-\frac{2D}{3}\ldots\ldots\right)x\right] = -\frac{1}{6}\left[e^{x}\left(x+\frac{2}{3}\right)-e^{-x}\left(x-\frac{2}{3}\right)\right]$$

$$= -\frac{x}{3}\left(\frac{e^{x}-e^{-x}}{2}\right)-\frac{2}{9}\left(\frac{e^{x}+e^{-x}}{2}\right) = -\frac{x}{3}\sinh x - \frac{2}{9}\cosh x$$

Hence the C.S. is $y = c_1e^{2x} + c_2e^{-2x} - \frac{x}{3}\sinh x - \frac{2}{9}\cosh x$.

Example 5. *Solve* $\frac{d^4y}{dx^4} - y = \cos x \cosh x$.

Sol. Given equation in symbolic form is $(D^4 - 1)y = \cos x \cosh x$

A.E. is $m^4 - 1 = 0$ or $(m^2 - 1)(m^2 + 1) = 0$ so that $m = \pm 1, \pm i$

$\therefore$ C.F. $= c_1e^{x} + c_2e^{-x} + e^{0x}(c_3\cos x + c_4\sin x) = c_1e^{x} + c_2e^{-x} + c_3\cos x + c_4\sin x$

$$\text{P.I.} = \frac{1}{D^4-1}\cos x\cosh x = \frac{1}{D^4-1}\cos x\left(\frac{e^{x}+e^{-x}}{2}\right)$$

$$= \frac{1}{2}\left[\frac{1}{D^4-1}e^{x}\cos x + \frac{1}{D^4-1}e^{-x}\cos x\right]$$

$$= \frac{1}{2}\left[e^{x}\frac{1}{(D+1)^4-1}\cos x + e^{-x}\frac{1}{(D-1)^4-1}\cos x\right]$$

$$= \frac{1}{2}\left[e^{x}\frac{1}{D^4+4D^3+6D^2+4D}\cos x + e^{-x}\frac{1}{D^4-4D^3+6D^2-4D}\cos x\right]$$

$$= \frac{1}{2}\left[e^{x}\frac{1}{(-1^2)^2+4D(-1^2)+6(-1^2)+4D}\cos x\right.$$

$$\left. + e^{-x}\frac{1}{(-1^2)^2-4D(-1^2)+6(-1^2)-4D}\cos x\right]$$

$$= \frac{1}{2}\left[e^{x}\frac{1}{-5}\cos x + e^{-x}\frac{1}{-5}\cos x\right] = -\frac{1}{5}\left(\frac{e^{x}+e^{-x}}{2}\right)\cos x = -\frac{1}{5}\cosh x\cos x$$

Hence the C.S. is $y = c_1e^x + c_2e^{-x} + c_3 \cos x + c_4 \sin x - \frac{1}{5} \cos x \cosh x$.

Example 6. *Solve* $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = xe^x \sin x$.

Sol. Given equation in symbolic form is $(D^2 - 2D + 1)y = xe^x \sin x$

A.E. is $m^2 - 2m + 1 = 0$ or $(m-1)^2 = 0$ so that $m = 1, 1$

$\therefore$ C.F. $= (c_1 + c_2x)e^x$

$$\text{P.I.} = \frac{1}{(D-1)^2} e^x . x \sin x = e^x . \frac{1}{(D+1-1)^2} x \sin x$$

$$= e^x \frac{1}{D^2} x \sin x = e^x \frac{1}{D} \int x \sin x \, dx$$

Integrating by parts,

$$\text{P.I.} = e^x \frac{1}{D}\left[x(-\cos x) - \int 1(-\cos x)\,dx\right] = e^x \frac{1}{D}(-x\cos x + \sin x)$$

$$= e^x \int (-x\cos x + \sin x)\,dx = e^x\left[-\left\{x \sin x - \int 1.\sin x\,dx\right\} - \cos x\right]$$

$$= e^x[-x \sin x - \cos x - \cos x] = -e^x(x \sin x + 2\cos x)$$

Hence the C.S. is $y = (c_1 + c_2x)e^x - e^x(x \sin x + 2 \cos x)$.

Example 7. *Solve* $\frac{d^2y}{dx^2} + y = cosec\ x$.

Sol. Given equation in symbolic form is $(D^2 + 1)y = \operatorname{cosec} x$

A.E. is $m^2 + 1 = 0 \Rightarrow m = \pm i$

$\therefore$ C.F. $= c_1 \cos x + c_2 \sin x$

$$\text{P.I.} = \frac{1}{D^2+1} \operatorname{cosec} x = \frac{1}{(D+i)(D-i)} \operatorname{cosec} x$$

$$= \frac{1}{2i}\left(\frac{1}{D-i} - \frac{1}{D+i}\right) \operatorname{cosec} x \qquad \text{(Partial Fractions)}$$

$$= \frac{1}{2i}\left(\frac{1}{D-i} \operatorname{cosec} x - \frac{1}{D+i} \operatorname{cosec} x\right)$$

Now $\frac{1}{D-i} \operatorname{cosec} x = e^{ix} \int \operatorname{cosec} x \, e^{-ix}\,dx$ $\left[\because \frac{1}{D-a} X = e^{ax} \int Xe^{-ax}\,dx\right]$

$$= e^{ix} \int \operatorname{cosec} x (\cos x - i \sin x)\,dx = e^{ix} \int (\cot x - i)\,dx$$

$$= e^{ix} (\log \sin x - ix)$$

Changing i to $-i$, we have $\frac{1}{D+i} \operatorname{cosec} x = e^{-ix} (\log \sin x + ix)$

$\therefore$ $$\text{P.I.} = \frac{1}{2i}[e^{ix}(\log \sin x - ix) - e^{-ix}(\log \sin x + ix)]$$

$$= \log \sin x \left(\frac{e^{ix} - e^{-ix}}{2i}\right) - x\left(\frac{e^{ix} + e^{-ix}}{2}\right)$$

$$= \log \sin x \,(\sin x) - x \cos x = \sin x \log \sin x - x \cos x$$

Hence the C.S. is $y = c_1 \cos x + c_2 \sin x + \sin x \log \sin x - x \cos x$.

Example 8. *Solve* $\dfrac{d^2 y}{dx^2} + a^2 y = \tan ax$.

Sol. Given equation in symbolic form is $(D^2 + a^2)y = \tan ax$

A.E. is $\quad m^2 + a^2 = 0 \quad \Rightarrow \quad m = \pm ia$

$\therefore \qquad$ C.F. $= c_1 \cos ax + c_2 \sin ax$

$$\text{P.I.} = \frac{1}{D^2 + a^2} \tan ax = \frac{1}{(D + ia)(D - ia)} \tan ax$$

$$= \frac{1}{2ia}\left[\frac{1}{D - ia} - \frac{1}{D + ia}\right] \tan ax \qquad \text{(Partial Fractions)}$$

$$= \frac{1}{2ia}\left[\frac{1}{D - ia} \tan ax - \frac{1}{D + ia} \tan ax\right]$$

Now $\dfrac{1}{D - ia} \tan ax = e^{iax} \displaystyle\int \tan ax \,.\, e^{-iax}\, dx$

$$= e^{iax} \int \tan ax\,(\cos ax - i \sin ax)\, dx = e^{iax} \int \left(\sin ax - i\,\frac{\sin^2 ax}{\cos ax}\right) dx$$

$$= e^{iax} \int \left(\sin ax - i\,\frac{1 - \cos^2 ax}{\cos ax}\right) dx = e^{iax} \int [\sin ax - i(\sec ax - \cos ax)]\, dx$$

$$= e^{iax}\left[-\frac{\cos ax}{a} - \frac{i}{a} \log(\sec ax + \tan ax) + i\,\frac{\sin ax}{a}\right]$$

$$= -\frac{1}{a}\, e^{iax}\, [(\cos ax - i \sin ax) + i \log(\sec ax + \tan ax)]$$

$$= -\frac{1}{a}\, e^{iax}\, [e^{-iax} + i \log(\sec ax + \tan ax)] = -\frac{1}{a}\, [1 + ie^{iax} \log(\sec ax + \tan ax)]$$

Changing i to $-i$, we have $\dfrac{1}{D + ia} \tan ax = -\dfrac{1}{a}\, [1 - ie^{-iax} \log(\sec ax + \tan ax)]$

$$\therefore \quad \text{P.I.} = \frac{1}{2ia}\left[-\frac{1}{a}\left\{1 + ie^{iax} \log(\sec ax + \tan ax)\right\} + \frac{1}{a}\left\{1 - ie^{-iax} \log(\sec ax + \tan ax)\right\}\right]$$

$$= -\frac{1}{a^2} \log(\sec ax + \tan ax)\left(\frac{e^{iax} + e^{-iax}}{2}\right) = -\frac{1}{a^2} \log(\sec ax + \tan ax) \,.\, \cos ax$$

Hence the C.S. is $y = c_1 \cos ax + c_2 \sin ax - \dfrac{1}{a^2} \cos ax \log(\sec ax + \tan ax)$.

TEST YOUR KNOWLEDGE

Solve the following differential equations :

1. $\dfrac{d^3y}{dx^3} + y = 3 + 5e^x$

2. $\dfrac{d^2y}{dx^2} - 4y = (1 + e^x)^2$

3. $\dfrac{d^2y}{dx^2} + 4\dfrac{dy}{dx} + 5y = -2\cosh x$

4. $\dfrac{d^2y}{dx^2} - 2\dfrac{dy}{dx} + 5y = \sin 3x$

5. $\dfrac{d^3y}{dx^3} + \dfrac{d^2y}{dx^2} + \dfrac{dy}{dx} + y = \sin 2x$

6. $\dfrac{d^3y}{dx^3} + y = \sin 3x - \cos^2 \dfrac{x}{2}$

7. $(D^2 - 4D + 3)y = \sin 3x \cos 2x$

8. $(D^2 - 3D + 2)y = 6e^{-3x} + \sin 2x$

9. $\dfrac{d^2y}{dx^2} + 4y = e^x + \sin 2x$

10. $\dfrac{d^3y}{dx^3} - 2\dfrac{d^2y}{dx^2} + 4\dfrac{dy}{dx} = e^{2x} + \sin 2x$

11. $\dfrac{d^2y}{dx^2} - 4y = x^2$

12. $\dfrac{d^3y}{dx^3} - \dfrac{d^2y}{dx^2} - 6\dfrac{dy}{dx} = 1 + x^2$

13. $\dfrac{d^2y}{dx^2} + \dfrac{dy}{dx} = x^2 + 2x + 4$

14. $\dfrac{d^2y}{dx^2} + y = e^{2x} + \cosh 2x + x^3$

15. $(D^2 - 3D + 2)y = 2e^x \cos \dfrac{x}{2}$

16. $\dfrac{d^2y}{dx^2} - 3\dfrac{dy}{dx} + 2y = xe^{3x} + \sin 2x$

17. $\dfrac{d^4y}{dx^4} - y = e^x \cos x$

18. $(D^2 - 2D)y = e^x \sin x$

19. $(D^2 + 4D + 8)y = 12e^{-2x} \sin x \sin 3x$

20. $\dfrac{d^2y}{dx^2} + 2y = x^2e^{3x} + e^x \cos 2x$

21. $(D^3 + 2D^2 + D)y = x^2e^{2x} + \sin^2 x$

22. $(D^2 - 4D + 4)y = 8x^2 e^{2x} \sin 2x$

23. $(D - 1)^2(D + 1)^2y = \sin^2 \dfrac{x}{2} + e^x + x$

24. $\dfrac{d^2y}{dx^2} + 4y = x \sin x$

25. $(D^2 - 1)y = x \sin x + (1 + x^2)e^x$

26. $\dfrac{d^2y}{dx^2} + a^2y = \sec ax$

27. $\dfrac{d^2y}{dx^2} + 4y = 4 \tan 2x$

28. $\dfrac{d^2y}{dx^2} + 3\dfrac{dy}{dx} + 2y = e^{e^x}$.

Answers

1. $y = c_1e^{-x} + e^{\frac{1}{2}x}\left(c_2 \cos \dfrac{\sqrt{3}}{2}x + c_3 \sin \dfrac{\sqrt{3}}{2}x\right) + 3 + \dfrac{5}{2}e^x$

2. $y = c_1e^{2x} + c_2e^{-2x} - \dfrac{1}{4} - \dfrac{2}{3}e^x + \dfrac{1}{4}xe^{2x}$

3. $y = e^{-2x}(c_1 \cos x + c_2 \sin x) - \dfrac{1}{10}e^x - \dfrac{1}{2}e^{-x}$

4. $y = e^x(c_1 \cos 2x + c_2 \sin 2x) + \dfrac{1}{26}(3 \cos 3x - 2 \sin 3x)$

5. $y = c_1 e^{-x} + c_2 \cos x + c_3 \sin x + \frac{1}{15}(2\cos 2x - \sin 2x)$

6. $y = c_1 e^{-x} + e^{\frac{1}{2}x}\left(c_2 \cos \frac{\sqrt{3}}{2}x + c_3 \sin \frac{\sqrt{3}}{2}x\right) + \frac{1}{730}(\sin 3x + 27\cos 3x) - \frac{1}{2} - \frac{1}{4}(\cos x - \sin x)$

7. $y = c_1 e^{x} + c_2 e^{3x} + \frac{1}{884}(10\cos 5x - 11\sin 5x) + \frac{1}{20}(\sin x + 2\cos x)$

8. $y = c_1 e^{x} + c_2 e^{2x} + \frac{3}{10}e^{-3x} + \frac{1}{20}(3\cos 2x - \sin 2x)$

9. $y = c_1 \cos 2x + c_2 \sin 2x + \frac{1}{5}e^{x} - \frac{x}{4}\cos 2x$

10. $y = c_1 + e^{x}(c_2 \cos \sqrt{3}x + c_3 \sin \sqrt{3}x) + \frac{1}{8}(e^{2x} + \sin 2x)$

11. $y = c_1 e^{2x} + c_2 e^{-2x} - \frac{1}{4}\left(x^2 + \frac{1}{2}\right)$

12. $y = c_1 + c_2 e^{3x} + c_3 e^{-2x} - \frac{1}{18}\left(x^3 - \frac{x^2}{2} + \frac{25}{6}x\right)$

13. $y = c_1 + c_2 e^{-x} + \frac{x^3}{3} + 4x$

14. $y = c_1 \cos x + c_2 \sin x + \frac{1}{5}e^{2x} + \frac{1}{5}\cosh 2x + x^3 - 6x$

15. $y = c_1 e^{x} + c_2 e^{2x} - \frac{8}{5}e^{x}\left(2\sin\frac{x}{2} + \cos\frac{x}{2}\right)$

16. $y = c_1 e^{x} + c_2 e^{2x} + \frac{1}{4}e^{3x}(2x - 3) + \frac{1}{20}(3\cos 2x - \sin 2x)$

17. $y = c_1 e^{x} + c_2 e^{-x} + c_3 \cos x + c_4 \sin x - \frac{1}{5}e^{x}\cos x$

18. $y = c_1 + c_2 e^{2x} - \frac{1}{2}e^{x}\sin x$

19. $y = e^{-2x}(c_1 \cos 2x + c_2 \sin 2x) + \frac{1}{2}e^{-2x}(3x \sin 2x + \cos 4x)$

20. $y = c_1 \cos \sqrt{2}x + c_2 \sin \sqrt{2}x + \frac{e^{3x}}{11}\left(x^2 - \frac{12}{11}x + \frac{50}{121}\right) + \frac{e^{x}}{17}(4\sin 2x - \cos 2x)$

21. $y = c_1 + (c_2 + c_3 x)e^{-x} + \frac{x}{2} + \frac{e^{2x}}{18}\left(x^2 - \frac{7}{3}x + \frac{11}{6}\right) + \frac{1}{100}(3\sin 2x + 4\cos 2x)$

22. $y = (c_1 + c_2 x)e^{2x} - e^{2x}[4x\cos 2x + (2x^2 - 3)\sin 2x]$

23. $y = (c_1 + c_2 x)e^{x} + (c_3 + c_4 x)e^{-x} + \frac{1}{2} - \frac{1}{8}\cos x + \frac{x^2}{8}e^{x} + x$

24. $y = c_1 \cos 2x + c_2 \sin 2x + \frac{1}{9}(3x \sin x - 2\cos x)$

25. $y = c_1 e^{x} + c_2 e^{-x} - \frac{1}{2}(x \sin x + \cos x) + \frac{1}{12}xe^{x}(2x^2 - 3x + 9)$

26. $y = c_1 \cos ax + c_2 \sin ax + \frac{1}{a}\left(x \sin ax + \cos ax \frac{\log \cos ax}{a}\right)$

27. $y = c_1 \cos 2x + c_2 \sin 2x - \cos 2x \log(\sec 2x + \tan 2x)$

28. $y = c_1 e^{-x} + c_2 e^{-2x} + e^{-2x} e^{e^{x}}$.

13.8. METHOD OF VARIATION OF PARAMETERS TO FIND P.I.

Consider the linear equation of **second order** with constant co-efficients

$$\frac{d^2y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = X \qquad ...(1)$$

Let its C.F. be $y = c_1 y_1 + c_2 y_2$ so that y_1 and y_2 satisfy the equation

$$\frac{d^2y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = 0 \qquad ...(2)$$

Now, let us assume that the P.I. of (1) is $y = uy_1 + vy_2$ ...(3)

where u and v are unknown functions of x.

Differentiating (3) w.r.t. x, we have $y' = uy_1' + vy_2' + u'y_1 + v'y_2 = uy_1' + vy_2'$ ...(4)

assuming that u, v satisfy the equation $u'y_1 + v'y_2 = 0$ ...(5)

Differentiating (4) w.r.t. x, we have $y'' = uy_1'' + u'y_1' + vy_2'' + v'y_2'$

Substituting the values of y, y' and y'' in (1), we get

$$(uy_1'' + u'y_1' + vy_2'' + v'y_2') + a_1(uy_1' + vy_2') + a_2(uy_1 + vy_1) = X$$

or $$u(y_1'' + a_1 y_1' + a_2 y_1) + v(y_2'' + a_1 y_2' + a_2 y_2) + u'y_1' + v'y_2' = X$$

or $$u'y_1' + v'y_2' = X \qquad ...(6)$$

since y_1 and y_2 satisfy (2).

Solving (5) and (6), we get $u' = \begin{vmatrix} 0 & y_2 \\ X & y_2' \end{vmatrix} \div \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = -\frac{y_2 X}{W}$

and $$v' = \begin{vmatrix} y_1 & 0 \\ y_1' & X \end{vmatrix} \div \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \frac{y_1 X}{W}$$

where $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$ is called the Wronskian of y_1, y_2.

Integrating $$u = -\int \frac{y_2 X}{W}\, dx, \quad v = \int \frac{y_1 X}{W}\, dx$$

Substituting in (3), the P.I. is known.

Note. As the solution is obtained by varying the arbitrary constants c_1, c_2 of the C.F., the method is known as *variation of parameters*.

Example. *Apply the method of variation of parameters to solve* $\frac{d^2y}{dx^2} + 4y = 4 \sec^2 2x$.

Sol. Given equation in symbolic form is $(D^2 + 4)y = 4 \sec^2 2x$

Its A.E. is $m^2 + 4 = 0$ so that $m = \pm 2i$

$\therefore$ C.F. is $y = c_1 \cos 2x + c_2 \sin 2x$

Here $y_1 = \cos 2x$, $y_2 = \sin 2x$ and $X = 4 \sec^2 2x$

$$\therefore \quad W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos 2x & \sin 2x \\ -2\sin 2x & 2\cos 2x \end{vmatrix} = 2$$

$$\text{P.I.} = -y_1 \int \frac{y_2 X}{W}\, dx + y_2 \int \frac{y_1 X}{W}\, dx$$

$$= -\cos 2x \int \frac{\sin 2x \,.\, 4\sec^2 2x}{2}\, dx + \sin 2x \int \frac{\cos 2x \,.\, 4 \sec^2 2x}{2}\, dx$$

$$= -2\cos 2x \int \sec 2x \tan 2x \, dx + 2 \sin 2x \int \sec 2x \, dx$$

$$= -2\cos 2x \, . \, \frac{\sec 2x}{2} + 2\sin 2x \, . \, \frac{1}{2} \log(\sec 2x + \tan 2x)$$

$$= -1 + \sin 2x \log(\sec 2x + \tan 2x)$$

Hence the C.S. is $y = c_1 \cos 2x + c_2 \sin 2x - 1 + \sin 2x \log(\sec 2x + \tan 2x)$.

TEST YOUR KNOWLEDGE

Solve by the method of variation of parameters :

1. $\dfrac{d^2y}{dx^2} + y = \operatorname{cosec} x.$

2. $\dfrac{d^2y}{dx^2} + y = \sec x$

3. $\dfrac{d^2y}{dx^2} + 4y = \tan 2x$

4. $\dfrac{d^2y}{dx^2} + y = x \sin x$

5. $y'' - 2y' + 2y = e^x \tan x.$

Answers

1. $y = c_1 \cos x + c_2 \sin x - x \cos x + \sin x \log \sin x$

2. $y = c_1 \cos x + c_2 \sin x + \cos x \log(\cos x) + x \sin x$

3. $y = c_1 \cos 2x + c_2 \sin 2x - \dfrac{1}{4} \cos 2x \log(\sec 2x + \tan 2x)$

4. $y = c_1 \cos x + c_2 \sin x + \dfrac{x}{4} \sin x - \dfrac{x^2}{4} \cos x$

5. $y = e^x (c_1 \cos x + c_2 \sin x) - e^x \cos x \log(\sec x + \tan x).$

13.9. CAUCHY'S HOMOGENEOUS LINEAR EQUATION

An equation of the form

$$x^n \frac{d^n y}{dx^n} + a_1 x^{n-1} \frac{d^{n-1}y}{dx^{n-1}} + a_2 x^{n-2} \frac{d^{n-2}y}{dx^{n-2}} + \ldots\ldots + a_{n-1} x \frac{dy}{dx} + a_n y = \mathrm{X} \qquad \ldots(i)$$

where a_i's are constants and X is a function of x, is called Cauchy's homogeneous linear equation.

Such equations can be reduced to linear differential equations with constant co-efficients by the substitution $\quad x = e^z \quad$ or $\quad z = \log x$

so that $$\frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{dz}{dx} = \frac{dy}{dz} \cdot \frac{1}{x} \quad \text{or} \quad x\frac{dy}{dx} = \frac{dy}{dz} = \mathrm{D}y, \text{ where } \mathrm{D} = \frac{d}{dz}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{1}{x} \cdot \frac{dy}{dz}\right) = -\frac{1}{x^2}\frac{dy}{dz} + \frac{1}{x} \cdot \frac{d^2y}{dz^2} \cdot \frac{dz}{dx}$$

$$= -\frac{1}{x^2}\frac{dy}{dz} + \frac{1}{x^2}\frac{d^2y}{dz^2} \qquad \left(\because \frac{dz}{dx} = \frac{1}{x}\right)$$

or $$x^2 \frac{d^2y}{dx^2} = \frac{d^2y}{dz^2} - \frac{dy}{dz} = \mathrm{D}^2y - \mathrm{D}y = \mathrm{D}(\mathrm{D}-1)y$$

Similarly, $x^3 \dfrac{d^3y}{dx^3} = D(D-1)(D-2)y$ and so on.

Substituting these values in equation (*i*), we get a linear differential equation with constant co-efficients, which can be solved by the methods already discussed.

ILLUSTRATIVE EXAMPLES

Example 1. *Solve* $x^3 \dfrac{d^3y}{dx^3} + 2x^2 \dfrac{d^2y}{dx^2} + 2y = 10\left(x + \dfrac{1}{x}\right)$.

Sol. Given equation is a Cauchy's homogeneous linear equation.

Put $x = e^z$ *i.e.*, $z = \log x$

so that $$x \frac{dy}{dx} = Dy,\ x^2 \frac{d^2y}{dx^2} = D(D-1)y$$

$$x^3 \frac{d^3y}{dx^3} = D(D-1)(D-2)y, \text{ where } D \equiv \frac{d}{dz}$$

Substituting these values in the given equation, it reduces to

$$[D(D-1)(D-2) + 2D(D-1) + 2]y = 10(e^z + e^{-z})$$

or $$(D^3 - D^2 + 2)y = 10(e^z + e^{-z})$$

which is a linear equation with constant co-efficients.

Its A.E. is $m^3 - m^2 + 2 = 0$ or $(m+1)(m^2 - 2m + 2) = 0$

$$\therefore \quad m = -1, \frac{2 \pm \sqrt{4-8}}{2} = -1, 1 \pm i$$

$$\therefore \quad \text{C.F.} = c_1 e^{-z} + e^z(c_2 \cos z + c_3 \sin z) = \frac{c_1}{x} + x[c_2 \cos(\log x) + c_3 \sin(\log x)]$$

$$\text{P.I.} = 10 \frac{1}{D^3 - D^2 + 2}(e^z + e^{-z}) = 10\left(\frac{1}{D^3 - D^2 + 2} e^z + \frac{1}{D^3 - D^2 + 2} e^{-z}\right)$$

$$= 10\left(\frac{1}{1^3 - 1^2 + 2} e^z + z \cdot \frac{1}{3D^2 - 2D} e^{-z}\right) = 10\left(\frac{1}{2} e^z + z \cdot \frac{1}{3(-1)^2 - 2(-1)} e^{-z}\right)$$

$$= 5e^z + 2ze^{-z} = 5x + \frac{2}{x} \log x$$

Hence the C.S. is $y = \dfrac{c_1}{x} + x[c_2 \cos(\log x) + c_3 \sin(\log x)] + 5x + \dfrac{2}{x} \log x$.

Example 2. *Solve* $x^2 \dfrac{d^2y}{dx^2} - x \dfrac{dy}{dx} - 3y = x^2 \log x$.

Sol. Given equation is a Cauchy's homogeneous linear equation.

Put $x = e^z$ *i.e.*, $z = \log x$ so that $x \dfrac{dy}{dx} = Dy,\ x^2 \dfrac{d^2y}{dx^2} = D(D-1)y$, where $D \equiv \dfrac{d}{dz}$.

Substituting these values in the given equation, it reduces to

$$[D(D-1)-D-3]y = ze^{2z} \quad \text{or} \quad (D^2-2D-3)y = ze^{2z}$$

which is a linear equation with constant co-efficients.

Its A.E. is $m^2 - 2m - 3 = 0$ or $(m-3)(m+1) = 0$

$\therefore$ $m = 3, -1$

$$\text{C.F.} = c_1 e^{3z} + c_2 e^{-z} = c_1 x^3 + \frac{c_2}{x}$$

$$\text{P.I.} = \frac{1}{D^2 - 2D - 3}(e^{2z}\,.\,z) = e^{2z}\frac{1}{(D+2)^2 - 2(D+2) - 3}z = e^{2z}\frac{1}{D^2 + 2D - 3}z$$

$$= e^{2z}\frac{1}{-3\left(1 - \frac{2D}{3} - \frac{D^2}{3}\right)}z = -\frac{1}{3}e^{2z}\left[1 - \left(\frac{2D}{3} + \frac{D^2}{3}\right)\right]^{-1} z$$

$$= -\frac{1}{3}e^{2z}\left[1 + \left(\frac{2D}{3} + \frac{D^2}{3}\right) + \ldots\ldots\right]z = -\frac{1}{3}e^{2z}\left(z + \frac{2}{3}\right) = -\frac{x^2}{3}\left(\log x + \frac{2}{3}\right)$$

Hence the C.S. is $y = c_1 x^3 + \dfrac{c_2}{x} - \dfrac{x^2}{3}\left(\log x + \dfrac{2}{3}\right)$.

13.10. LEGENDRE'S LINEAR EQUATION

An equation of the form

$$(a+bx)^n \frac{d^n y}{dx^n} + a_1 (a+bx)^{n-1}\frac{d^{n-1}y}{dx^{n-1}} + \ldots\ldots + a_{n-1}(a+bx)\frac{dy}{dx} + a_n y = X \quad \ldots$$

where a_i's are constants and X is a function of x, is called Legendre's linear equation.

Such equations can be reduced to linear differential equations with constant co-efficien by the substitution $a + bx = e^z$ *i.e.,* $z = \log(a+bx)$ so that $\dfrac{dy}{dx} = \dfrac{dy}{dz}\cdot\dfrac{dz}{dx} = \dfrac{b}{a+bx}\dfrac{dy}{dz}$

or $$(a+bx)\frac{dy}{dx} = b\frac{dy}{dz} = b\,Dy, \quad \text{where } D = \frac{d}{dz}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{b}{a+bx}\frac{dy}{dz}\right) = -\frac{b^2}{(a+bx)^2}\frac{dy}{dz} + \frac{b}{a+bx}\cdot\frac{d^2y}{dz^2}\cdot\frac{dz}{dx}$$

$$= -\frac{b^2}{(a+bx)^2}\frac{dy}{dz} + \frac{b}{a+bx}\frac{d^2y}{dz^2}\cdot\frac{b}{a+bx} = \frac{b^2}{(a+bx)^2}\left(\frac{d^2y}{dz^2} - \frac{dy}{dz}\right)$$

or $$(a+bx)^2\frac{d^2y}{dx^2} = b^2(D^2y - Dy) = b^2 D(D-1)y$$

Similarly, $(a+bx)^3\dfrac{d^3y}{dx^3} = b^3 D(D-1)(D-2)y$.

Substituting these values in equation (i), we get a linear differential equation wi constant co-efficients, which can be solved by the methods already discussed.

Example. *Solve* $(3x + 2)^2 \dfrac{d^2y}{dx^2} + 3(3x + 2)\dfrac{dy}{dx} - 36y = 3x^2 + 4x + 1.$

Sol. Given equation is a Legendre's linear equation.

Put $\quad 3x + 2 = e^z \quad$ *i.e.,* $\quad z = \log(3x + 2) \quad$ so that $\quad (3x + 2)\dfrac{dy}{dx} = 3\text{D}y.$

$$(3x + 2)^2 \frac{d^2y}{dx^2} = 3^2\,\text{D}(\text{D} - 1)y, \quad \text{where } \text{D} \equiv \frac{d}{dz}.$$

Substituting these values in the given equation, it reduces to

$$[3^2\,\text{D}(\text{D} - 1) + 3.3\text{D} - 36]y = 3\left(\frac{e^z - 2}{3}\right)^2 + 4\left(\frac{e^z - 2}{3}\right) + 1$$

or
$$9(\text{D}^2 - 4)y = \frac{1}{3}e^{2z} - \frac{1}{3}$$

or
$$(\text{D}^2 - 4)y = \frac{1}{27}(e^{2z} - 1)$$

which is a linear equation with constant co-efficients.

Its A.E. is $\quad m^2 - 4 = 0 \quad \therefore \quad m = \pm 2$

$$\text{C.F.} = c_1e^{2z} + c_2e^{-2z} = c_1(3x + 2)^2 + c_2(3x + 2)^{-2}$$

$$\text{P.I.} = \frac{1}{27}\cdot\frac{1}{\text{D}^2 - 4}(e^{2z} - 1) = \frac{1}{27}\left[\frac{1}{\text{D}^2 - 4}e^{2z} - \frac{1}{\text{D}^2 - 4}e^{0z}\right]$$

$$= \frac{1}{27}\left[z\,.\,\frac{1}{2\text{D}}e^{2z} - \frac{1}{0 - 4}e^{0z}\right]$$

$$= \frac{1}{27}\left[\frac{z}{4}e^{2z} + \frac{1}{4}\right] = \frac{1}{108}(ze^{2z} + 1) = \frac{1}{108}[(3x + 2)^2\log(3x + 2) + 1]$$

Hence the C.S. is $y = c_1(3x + 2)^2 + c_2(3x + 2)^{-2} + \dfrac{1}{108}[(3x + 2)^2\log(3x + 2) + 1].$

TEST YOUR KNOWLEDGE

Solve :

1. $x^2\dfrac{d^2y}{dx^2} + 9x\dfrac{dy}{dx} + 25y = 50$

2. $x^2\dfrac{d^2y}{dx^2} - 2y = x^2 + \dfrac{1}{x}$

3. $x^2\dfrac{d^2y}{dx^2} + 2x\dfrac{dy}{dx} - 20y = (x + 1)^2$

4. $x^2\dfrac{d^3y}{dx^3} - 4x\dfrac{d^2y}{dx^2} + 6\dfrac{dy}{dx} = 4$

[**Hint.** Multiply throughout by x]

5. (*i*) $x^4\dfrac{d^3y}{dx^3} + 2x^3\dfrac{d^2y}{dx^2} - x^2\dfrac{dy}{dx} + xy = 1.$ (*ii*) $x^2\dfrac{d^2y}{dx^2} - 4x\dfrac{dy}{dx} + 6y = x^2.$

6. The radial displacement u in a rotating disc at a distance r from the axis is given by $r^2\dfrac{d^2u}{dr^2} + r\dfrac{du}{dr} - u + kr^3 = 0$, where k is a constant. Solve the equation under the conditions $u = 0$ when $r = 0$, $u = 0$ when $r = a$.

7. $x^2 \dfrac{d^2y}{dx^2} - x\dfrac{dy}{dx} + y = \log x$

8. $x^2 \dfrac{d^2y}{dx^2} - x\dfrac{dy}{dx} + 2y = x\log x$

9. $x^2 \dfrac{d^2y}{dx^2} + 2x\dfrac{dy}{dx} - 12y = x^3\log x$

10. $x^2 \dfrac{d^2y}{dx^2} - 2x\dfrac{dy}{dx} - 4y = x^2 + 2\log x$

11. $x^2 \dfrac{d^2y}{dx^2} - 3x\dfrac{dy}{dx} + 5y = \sin(\log x)$

12. $x^3 \dfrac{d^3y}{dx^3} + 3x^2\dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} + 8y = 65\cos(\log x)$

13. $x^2 \dfrac{d^2y}{dx^2} - 3x\dfrac{dy}{dx} + 5y = x^2\sin(\log x)$

14. $x^2 \dfrac{d^2y}{dx^2} - 3x\dfrac{dy}{dx} + y = \dfrac{\log x\sin(\log x) + 1}{x}$

15. $\dfrac{d^2y}{dx^2} + \dfrac{1}{x}\cdot\dfrac{dy}{dx} = \dfrac{12\log x}{x^2}$

16. $(1+x)^2 \dfrac{d^2y}{dx^2} + (1+x)\dfrac{dy}{dx} + y = 4\cos\log(1+x)$

17. $(1+2x)^2 \dfrac{d^2y}{dx^2} - 6(1+2x)\dfrac{dy}{dx} + 16y = 8(1+2x)^2$

18. $(2x+3)^2 \dfrac{d^2y}{dx^2} - 2(2x+3)\dfrac{dy}{dx} - 12y = 6x.$

Answers

1. $y = x^{-4}[c_1\cos(3\log x) + c_2\sin(3\log x)] + 2$

2. $y = c_1x^2 + \dfrac{c_2}{x} + \dfrac{1}{3}\left(x^2 - \dfrac{1}{x}\right)\log x$

3. $y = c_1x^4 + c_2x^{-5} - \dfrac{x^2}{14} - \dfrac{x}{9} - \dfrac{1}{20}$

4. $y = c_1 + c_2x^3 + c_3x^4 + \dfrac{2}{3}x$

5. (*i*) $y = (c_1 + c_2\log x)x + c_3x^{-1} + \dfrac{1}{4x}\log x$ (*ii*) $y = c_1x^2 + c_2x^3 - x^2\log x$

6. $u = \dfrac{kr}{8}(a^2 - r^2)$

7. $y = (c_1 + c_2\log x)x + \log x + 2$

8. $y = x[c_1\cos(\log x) + c_2\sin(\log x)] + x\log x$

9. $y = c_1x^3 + c_2x^{-4} + \dfrac{x^3}{98}\log x(7\log x - 2)$

10. $y = c_1x^{-1} + c_2x^4 - \dfrac{x^2}{6} - \dfrac{1}{2}\log x + \dfrac{3}{8}$

11. $y = x^2[c_1\cos(\log x) + c_2\sin(\log x)] + \dfrac{1}{8}[\sin(\log x) + \cos(\log x)]$

12. $y = c_1x^{-2} + x[c_2\cos(\sqrt{3}\log x) + c_3\sin(\sqrt{3}\log x)] + 8\cos(\log x) - \sin(\log x)$

13. $y = x^2[c_1\cos(\log x) + c_2\sin(\log x)] - \dfrac{1}{2}x^2\log x\cos(\log x)$

14. $y = x^2[c_1\cosh(\sqrt{3}\log x) + c_2\sinh(\sqrt{3}\log x)]$

$$+ \frac{1}{61x}\left[\log x\{5\sin(\log x) + 6\cos(\log x)\} + \frac{2}{61}\{27\sin(\log x) + 191\cos(\log x)\}\right] + \frac{1}{6x}$$

15. $y = c_1\log x + c_2 + 2(\log x)^3$

16. $y = c_1\cos[\log(1+x)] + c_2\sin[\log(1+x)] + 2\log(1+x) + \sin[\log(1+x)]$

17. $y = (1+2x)^2[c_1 + c_2\log(1+2x) + \{\log(1+2x)\}^2]$

18. $y = c_1(2x+3)^{-1} + c^2(2x+3)^3 - \frac{3}{4}(2x+3) + 3.$

13.11. SIMULTANEOUS LINEAR EQUATIONS WITH CONSTANT CO-EFFICIENTS

Now we discuss differential equations in which there is one independent variable and two or more than two dependent variables. Such equations are called *simultaneous linear equations*. To solve such equations completely, we must have as many simultaneous equations as the number of dependent variables. Here, we shall consider simultaneous linear equations with constant co-efficients only.

Let x, y be the two dependent variables and t the independent variable. Consider the simultaneous equations

$$f_1(\mathrm{D})x + f_2(\mathrm{D})y = \mathrm{T}_1 \quad ...(1) \quad \text{and} \quad \phi_1(\mathrm{D})x + \phi_2(\mathrm{D})y = \mathrm{T}_2 \quad ...(2)$$

where $\mathrm{D} \equiv \dfrac{d}{dt}$ and T_1, T_2 are functions of t.

To eliminate y, operating on both sides of (1) by $\phi_2(\mathrm{D})$ and on both sides of (2) by $f_2(\mathrm{D})$ and subtracting, we get

$$[f_1(\mathrm{D})\,\phi_2(\mathrm{D}) - \phi_1(\mathrm{D})\,f_2(\mathrm{D})]\,x = \phi_2(\mathrm{D})\mathrm{T}_1 - f_2(\mathrm{D})\mathrm{T}_2 \quad \text{or} \quad f(\mathrm{D})\,x = \mathrm{T}$$

which is a linear equation in x and t and can be solved by the methods already discussed. Substituting the value of x in either (1) or (2), we get the value of y.

Note. We can also eliminate x to get a linear equation in y and t.

ILLUSTRATIVE EXAMPLES

Example 1. *Solve* $\dfrac{dx}{dt} + 4x + 3y = t,\ \dfrac{dy}{dt} + 2x + 5y = e^t.$

Sol. Writing D for $\dfrac{d}{dt}$, the given equations become $(\mathrm{D} + 4)x + 3y = t$ ...(1)

and $2x + (\mathrm{D} + 5)y = e^t$ ...(2)

To eliminate y, operating on both sides of (1) by (D + 5) and on both sides of (2) by 3 and subtracting, we get

$$[(\mathrm{D} + 4)(\mathrm{D} + 5) - 6]x = (\mathrm{D} + 5)t - 3e^t$$

or $$(\mathrm{D}^2 + 9\mathrm{D} + 14)x = 1 + 5t - 3e^t$$

Its A.E. is $$m^2 + 9m + 14 = 0$$

or $$(m + 2)(m + 7) = 0 \quad \therefore \quad m = -2, -7$$

$$\text{C.F.} = c_1e^{-2t} + c_2e^{-7t}$$

$$\text{P.I.} = \frac{1}{\mathrm{D}^2 + 9\mathrm{D} + 14}(1 + 5t - 3e^t)$$

$$= \frac{1}{\mathrm{D}^2 + 9\mathrm{D} + 14}e^{0t} + 5\frac{1}{\mathrm{D}^2 + 9\mathrm{D} + 14}t - 3\frac{1}{\mathrm{D}^2 + 9\mathrm{D} + 14}e^t$$

$$= \frac{1}{0^2 + 9(0) + 14}e^{0t} + 5\,.\,\frac{1}{14\left(1 + \dfrac{9\mathrm{D}}{14} + \dfrac{\mathrm{D}^2}{14}\right)}t - 3\frac{1}{1^2 + 9(1) + 14}e^t$$

$$= \frac{1}{14} + \frac{5}{14}\left[1 + \left(\frac{9\mathrm{D}}{14} + \frac{\mathrm{D}^2}{14}\right)\right]^{-1} t - \frac{1}{8}e^t = \frac{1}{14} + \frac{5}{14}\left[1 - \left(\frac{9\mathrm{D}}{14} + \frac{\mathrm{D}^2}{14}\right) + ...\right]t - \frac{1}{8}e^t$$

$$= \frac{1}{14} + \frac{5}{14}\left(t - \frac{9}{14}\right) - \frac{1}{8}e^t = \frac{5}{14}t - \frac{31}{196} - \frac{1}{8}e^t$$

$\therefore$ $$x = c_1e^{-2t} + c_2e^{-7t} + \frac{5}{14}t - \frac{31}{196} - \frac{1}{8}e^t$$

Now $$\frac{dx}{dt} = -2c_1e^{-2t} - 7c_2e^{-7t} + \frac{5}{14} - \frac{1}{8}e^t$$

Substituting the values of x and $\frac{dx}{dt}$ in (1), we have

$$3y = t - \frac{dx}{dt} - 4x$$

$$= t + 2c_1e^{-2t} + 7c_2e^{-7t} - \frac{5}{14} + \frac{1}{8}e^t - 4c_1e^{-2t} - 4c_2e^{-7t} - \frac{10}{7}t + \frac{31}{49} + \frac{1}{2}e^t$$

$\therefore$ $$y = \frac{1}{3}\left[-2c_1e^{-2t} + 3c_2e^{-7t} - \frac{3}{7}t + \frac{27}{98} + \frac{5}{8}e^t\right]$$

Hence $$x = c_1e^{-2t} + c_2e^{-7t} + \frac{5}{14}t - \frac{31}{196} - \frac{1}{8}e^t$$

$$y = -\frac{2}{3}c_1e^{-2t} + c_2e^{-7t} - \frac{1}{7}t + \frac{9}{98} + \frac{5}{24}e^t.$$

Example 2. *Solve* $\frac{d^2x}{dt^2} + 4x + 5y = t^2, \frac{d^2y}{dt^2} + 5x + 4y = t + 1.$

Sol. Writing D for $\frac{d}{dt}$, the given equations become $(D^2 + 4)x + 5y = t^2$ ...(1)

and $5x + (D^2 + 4)y = t + 1$ ...(2)

To eliminate y, operating on both sides of (1) by $(D^2 + 4)$ and on both sides of (2) by 5 and subtracting, we get

$$[(D^2 + 4)^2 - 25]x = (D^2 + 4)t^2 - 5(t + 1)$$

or $$(D^4 + 8D^2 - 9)x = 2 + 4t^2 - 5t - 5 = 4t^2 - 5t - 3$$

Its A.E. is $$m^4 + 8m^2 - 9 = 0$$

or $$(m^2 + 9)(m^2 - 1) = 0 \quad \therefore \quad m = \pm 1, \pm 3i$$

$$\text{C.F.} = c_1e^t + c_2e^{-t} + c_3\cos 3t + c_4\sin 3t$$

$$\text{P.I.} = \frac{1}{D^4 + 8D^2 - 9}(4t^2 - 5t - 3) = \frac{1}{-9\left(1 - \frac{8D^2}{9} - \frac{D^4}{9}\right)}(4t^2 - 5t - 3)$$

$$= -\frac{1}{9}\left[1 - \left(\frac{8D^2}{9} + \frac{D^4}{9}\right)\right]^{-1}(4t^2 - 5t - 3)$$

$$= -\frac{1}{9}\left[1 + \left(\frac{8D^2}{9} + \frac{D^4}{9}\right) + \ldots\ldots\right](4t^2 - 5t - 3)$$

$$= -\frac{1}{9}\left[4t^2 - 5t - 3 + \frac{8}{9}(8)\right] = -\frac{1}{9}\left(4t^2 - 5t + \frac{37}{9}\right)$$

$\therefore$ $$x = c_1e^t + c_2e^{-t} + c_3 \cos 3t + c_4 \sin 3t - \frac{4}{9}t^2 + \frac{5}{9}t - \frac{37}{81}$$

Now $$\frac{dx}{dt} = c_1e^t - c_2e^{-t} - 3c_3 \sin 3t + 3c_4 \cos 3t - \frac{8}{9}t + \frac{5}{9}$$

$$\frac{d^2x}{dt^2} = c_1e^t + c_2e^{-t} - 9c_3 \cos 3t - 9c_4 \sin 3t - \frac{8}{9}$$

Substituting the values of x and $\frac{d^2x}{dt^2}$ in (1), we have

$$5y = t^2 - 4x - \frac{d^2x}{dt^2}$$

$$= t^2 - 4c_1e^t - 4c_2e^{-t} - 4c_3 \cos 3t - 4c_4 \sin 3t$$

$$+ \frac{16}{9}t^2 - \frac{20}{9}t + \frac{148}{81} - c_1e^t - c_2e^{-t} + 9c_3 \cos 3t + 9c_4 \sin 3t + \frac{8}{9}$$

$\therefore$ $$y = \frac{1}{5}\left[-5c_1e^t - 5c_2e^{-t} + 5c_3 \cos 3t + 5c_4 \sin 3t + \frac{25}{9}t^2 - \frac{20}{9}t + \frac{220}{81}\right]$$

Hence $$x = c_1e^t + c_2e^{-t} + c_3 \cos 3t + c_4 \sin 3t - \frac{1}{9}\left(4t^2 - 5t + \frac{37}{9}\right)$$

$$y = -c_1e^t - c_2e^{-t} + c_3 \cos 3t + c_4 \sin 3t + \frac{1}{9}\left(5t^2 - 4t + \frac{44}{9}\right).$$

Example 3. *Solve the simultaneous equations :*

$$t\frac{dx}{dt} + y = 0,\ t\frac{dy}{dt} + x = 0 \text{ given } x(1) = 1,\ y(-1) = 0.$$

Sol. The given equations are $$t\frac{dx}{dt} + y = 0 \qquad ...(1)$$

$$t\frac{dy}{dt} + x = 0 \qquad ...(2)$$

Differentiating (1) w.r.t. t, we have

$$t\frac{d^2x}{dt^2} + \frac{dx}{dt} + \frac{dy}{dt} = 0$$

Multiplying throughout by t

$$t^2\frac{d^2x}{dt^2} + t\frac{dx}{dt} + t\frac{dy}{dt} = 0$$

or $$t^2\frac{d^2x}{dt^2} + t\frac{dx}{dt} - x = 0 \qquad \text{[Using (2)] ...(3)}$$

which is Cauchy's homogeneous linear equation.

Putting $t = e^u$ *i.e.*, $u = \log t$, so that $t\frac{d}{dt} = \frac{d}{du} = \text{D}$, equation (3) becomes

$$[\text{D}(\text{D} - 1) + \text{D} - 1]x = 0 \quad \text{or} \quad (\text{D}^2 - 1)\,x = 0$$

Its A.E. is $m^2 - 1 = 0$, whence $m = \pm 1$

$$\therefore \qquad x = c_1 e^u + c_2 e^{-u} = c_1 t + \frac{c_2}{t} \qquad ...(4)$$

From (1),
$$y = -t \frac{dx}{dt} = -t\left(c_1 - \frac{c_2}{t^2}\right) = -c_1 t + \frac{c_2}{t} \qquad ...(5)$$

Since $x(1) = 1$, $\therefore$ from (4), we have $1 = c_1 + c_2$

Also $y(-1) = 0$, $\therefore$ from (5), we have $0 = c_1 - c_2$

Solving
$$c_1 = c_2 = \frac{1}{2}$$

Hence
$$x = \frac{1}{2}\left(t + \frac{1}{t}\right), y = \frac{1}{2}\left(-t + \frac{1}{t}\right).$$

Example 4. *Solve the following simultaneous equations :*

$$\frac{dx}{dt} = 2y, \quad \frac{dy}{dt} = 2z, \quad \frac{dz}{dt} = 2x.$$

Sol. The given equations are

$$\frac{dx}{dt} = 2y \quad ...(1) \qquad \frac{dy}{dt} = 2z \quad ...(2) \qquad \frac{dz}{dt} = 2x \quad ...(3)$$

Differentiating (1) w.r.t. t,
$$\frac{d^2x}{dt^2} = 2\frac{dy}{dt} = 2(2z) \qquad \text{[Using (2)]}$$

Differentiating again w.r.t. t,
$$\frac{d^3x}{dt^3} = 4\frac{dz}{dt} = 4(2x)$$

or
$$(D^3 - 8)x = 0, \qquad \text{where } D \equiv \frac{d}{dt}$$

Its A.E. is
$$m^3 - 8 = 0 \quad \text{or} \quad (m-2)(m^2 + 2m + 4) = 0$$

whence
$$m = 2, -1 \pm i\sqrt{3}$$

$$\therefore \qquad x = c_1 e^{2t} + c_2 e^{-t} \cos(\sqrt{3}\,t - c_3)$$

From (1),
$$y = \frac{1}{2}\frac{dx}{dt}$$
$$= \frac{1}{2}[2c_1 e^{2t} - c_2 e^{-t}\cos(\sqrt{3}t - c_3) - c_2\sqrt{3}e^{-t}\sin(\sqrt{3}t - c_3)]$$
$$= c_1 e^{2t} + c_2 e^{-t}\left[\cos\frac{2\pi}{3}\cos(\sqrt{3}t - c_3) - \sin\frac{2\pi}{3}\sin(\sqrt{3}t - c_3)\right]$$
$$= c_1 e^{2t} + c_2 e^{-t}\cos\left(\sqrt{3}t - c_3 + \frac{2\pi}{3}\right)$$

From (2),
$$z = \frac{1}{2}\frac{dy}{dt}$$
$$= \frac{1}{2}\left[2c_1 e^{2t} - c_2 e^{-t}\cos\left(\sqrt{3}t - c_3 + \frac{2\pi}{3}\right) - c_2\sqrt{3}\,e^{-t}\sin\left(\sqrt{3}t - c_3 + \frac{2\pi}{3}\right)\right]$$

$$= c_1 e^{2t} + c_2 e^{-t} \left[\cos \frac{2\pi}{3} \cos \left(\sqrt{3}t - c_3 + \frac{2\pi}{3} \right) - \sin \frac{2\pi}{3} \sin \left(\sqrt{3}t - c_3 + \frac{2\pi}{3} \right) \right]$$

$$= c_1 e^{2t} + c_2 e^{-t} \cos \left(\sqrt{3}t - c_3 + \frac{4\pi}{3} \right).$$

TEST YOUR KNOWLEDGE

Solve the following simultaneous equations :

1. $\frac{dx}{dt} = 7x - y,\ \frac{dy}{dt} = 2x + 5y.$

2. $\frac{dx}{dt} + y = \sin t,\ \frac{dy}{dt} + x = \cos t$; given that $x = 2$ and $y = 0$ when $t = 0$.

3. $\frac{dx}{dt} + 5x - 2y = t,\ \frac{dy}{dt} + 2x + y = 0$; given that $x = y = 0$ when $t = 0$.

4. $(D + 1)x + (2D + 1)y = e^t,\ (D - 1)x + (D + 1)y = 1.$

5. $\frac{dx}{dt} + 2x + 3y = 0,\ 3x + \frac{dy}{dt} + 2y = 2e^{2t}.$

6. $(D - 1)x + Dy = 2t + 1,\ (2D + 1)x + 2Dy = t.$

7. $\frac{d^2x}{dt^2} - 3x - 4y = 0,\ \frac{d^2y}{dt^2} + x + y = 0.$

8. $\frac{d^2x}{dt^2} - \frac{dy}{dt} = 2x + 2t,\ \frac{dx}{dt} + 4\frac{dy}{dt} = 3y.$

9. $\frac{d^2y}{dt^2} + \frac{dy}{dt} - 2y = \sin t,\ \frac{dx}{dt} + x - 3y = 0.$

10. $\frac{d^2x}{dt^2} + y = \sin t,\ \frac{d^2y}{dt^2} + x = \cos t$

11. A mechanical system with two degrees of freedom satisfies the equations

$$2\frac{d^2x}{dt^2} + 3\frac{dy}{dt} = 4,\ 2\frac{d^2y}{dt^2} - 3\frac{dx}{dt} = 0$$

Obtain expressions for x and y in terms of t, given $x, y, \frac{dx}{dt}, \frac{dy}{dt}$ all vanish at $t = 0$.

Answers

1. $x = e^{6t}(c_1 \cos t + c_2 \sin t),\ y = e^{6t}[(c_1 - c_2)\cos t + (c_1 + c_2)\sin t]$

2. $x = e^t + e^{-t},\ y = e^{-t} - e^t + \sin t$

3. $x = -\frac{1}{27}(1 + 6t)e^{-3t} + \frac{1}{27}(1 + 3t),\ y = -\frac{2}{27}(2 + 3t)e^{-3t} + \frac{2}{27}(2 - 3t)$

4. $x = \frac{1}{2}[e^t - 1 - ac_1e^{at} + bc_2e^{bt}],\ y = c_1e^{at} + c_2e^{bt} + \frac{1}{2}$, where $a = \frac{1}{2}(3 + \sqrt{17}),\ b = \frac{1}{2}(3 - \sqrt{17})$

5. $x = c_1e^t + c_2e^{-5t} - \frac{6}{7}e^{2t},\ y = c_2e^{-5t} - c_1e^t + \frac{8}{7}e^{2t}$

6. $x = -t - \frac{2}{3},\ y = \frac{1}{2}t^2 + \frac{4}{3}t + c$

7. $x = (c_1 + c_2t)e^{-t} + (c_3 + c_4t)e^t,\ y = -\frac{1}{2}[c_1 + c_2(1 + t)]e^{-t} + \frac{1}{2}[c_4(1 - t) - c_3]e^t$

8. $x = (c_1 + c_2t)e^t + c_3e^{-\frac{3}{2}t} - t,\ y = [c_2(3 - t) - c_1]e^t - \frac{1}{6}c_3e^{-\frac{3}{2}t}$

9. $x = \frac{3}{2} c_1 e^t - 3c_2 e^{-2t} + c_3 e^{-t} + \frac{3}{10} e^t (\cos t - 2 \sin t),\ y = c_1 e^t + c_2 e^{-2t} - \frac{1}{10} (\cos t + 3 \sin t)$

10. $x = c_1 e^t + c_2 e^{-t} + c_3 \cos t + c_4 \sin t + \frac{t}{4} (\sin t - \cos t)$

$y = -c_1 e^t - c_2 e^{-t} + c_3 \cos t + c_4 \sin t + \frac{1}{4} (2 + t)(\sin t - \cos t).$

11. $x = \frac{8}{9}\left(1 - \cos \frac{3t}{2}\right),\ y = \frac{4}{3} t - \frac{8}{9} \sin \frac{3t}{2}.$

13.12. TOTAL DIFFERENTIAL EQUATIONS

An ordinary differential equation of the form $Pdx + Qdy + Rdz = 0$ ...(1)
where P, Q, R are functions of x, y, z is called the **total differential equation.**

The necessary and sufficient condition for integrability of (1) is

$$P\left(\frac{\partial Q}{\partial z} - \frac{\partial R}{\partial y}\right) + Q\left(\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z}\right) + R\left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right) = 0.$$

13.13. METHOD FOR SOLVING $\mathbf{Pdx + Qdy + Rdz = 0}$

Method I. Solution by inspection. When the condition of integrability is satisfied, it may be possible that by rearranging the terms, it becomes exact and the solution is found readily.

ILLUSTRATIVE EXAMPLES

Example 1. *Solve* : $(yz + 2x)\, dx + (zx - 2z)\, dy + (xy - 2y)\, dz = 0.$

Sol. Here $P = yz + 2x,\ Q = zx - 2z,\ R = xy - 2y$

$$\therefore \quad P\left(\frac{\partial Q}{\partial z} - \frac{\partial R}{\partial y}\right) + Q\left(\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z}\right) + R\left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right)$$

$$= (yz + 2x)(x - 2 - x + 2) + (zx - 2z)(y - y) + (xy - 2y)(z - z) = 0$$

Thus the condition of integrability is satisfied.

Regrouping the terms, we can write the given equation as

$$(yzdx + zxdy + xydz) + 2xdx - 2(zdy + ydz) = 0$$

Integrating, we get $xyz + x^2 - 2yz = c$ which is the required solution.

Example 2. *Solve* $(x + z)^2\, dy + y^2\, (dx + dz) = 0.$

Sol. The given equation can be written as $y^2dx + (x + z)^2\, dy + y^2dz = 0$

Here $P = y^2,\ Q = (x + z)^2,\ R = y^2$

$$\therefore \quad P\left(\frac{\partial Q}{\partial z} - \frac{\partial R}{\partial y}\right) + Q\left(\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z}\right) + R\left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right)$$

$$= y^2[2(x + z) - 2y] + (x + z)^2\,(0 - 0) + y^2(2y - 2(x + z))$$

$$= y^2(2x + 2z - 2y + 2y - 2x - 2z) = 0$$

Thus the condition of integrability is satisfied.

The given equation can be written as $\frac{dy}{y^2} + \frac{dx + dz}{(x+z)^2} = 0$

Integrating, $\quad -\frac{1}{y} - \frac{1}{x+z} = \frac{1}{c} \quad$ or $\quad -\frac{x+y+z}{y(x+z)} = \frac{1}{c}$

or $\quad c(x + y + z) + y(x + z) = 0$ which is the required solution.

Method II. One variable regarded as constant. In case the condition of integrability is satisfied and any two terms, say $\mathrm{P}dx + \mathrm{Q}dy = 0$ can be easily solved, then the third variable z is taken as constant, so that $dz = 0$.

Solve the equation $\mathrm{P}dx + \mathrm{Q}dy = 0$ by any of the methods discussed in chapter 11. Let the solution be $u = c$. Replace the arbitrary constant c by any arbitrary function of z, say $\phi(z)$.

Thus $u = \phi(z)$ is the solution of $\mathrm{P}dx + \mathrm{Q}dy = 0$, where $\phi(z)$ is a function of z only.

Differentiate $u = \phi(z)$ w.r.t. x, y, z and compare the result with the given equation to determine $\phi(z)$.

Example 3. *Solve $yz\,dx - 2xz\,dy + (xy - zy^3)\,dz = 0$.*

Sol. Here $\quad \mathrm{P} = yz,\ \mathrm{Q} = -2xz,\ \mathrm{R} = xy - zy^3$

$$\therefore \quad \mathrm{P}\left(\frac{\partial \mathrm{Q}}{\partial z} - \frac{\partial \mathrm{R}}{\partial y}\right) + \mathrm{Q}\left(\frac{\partial \mathrm{R}}{\partial x} - \frac{\partial \mathrm{P}}{\partial z}\right) + \mathrm{R}\left(\frac{\partial \mathrm{P}}{\partial y} - \frac{\partial \mathrm{Q}}{\partial x}\right)$$

$$= yz(-2x - x - 3y^2z) - 2xz(y - y) + (xy - zy^3)(z + 2z)$$

$$= yz(-3x + 3y^2z) + 3z(xy - zy^3) = 0.$$

Thus the condition of integrability is satisfied.

Considering z as constant so that $dz = 0$, the given equation becomes

$$yzdx - 2xzdy = 0$$

or $$\frac{dx}{x} - 2\frac{dy}{y} = 0$$

Integrating $\quad \log x - 2\log y = \log c$

or $$\frac{x}{y^2} = c \quad \Rightarrow \quad x = cy^2$$

Replacing c by $\phi(z)$, we have $x = y^2\phi(z) \quad$ or $\quad x - y^2\,\phi(z) = 0 \qquad$ ...(1)

Differentiating (1) w.r.t. x, y, z, we get $1dx - 2y\phi(z)\,dy - y^2\phi'(z)\,dz = 0 \qquad$ ...(2)

Comparing (2) with the given equation, we get $\frac{yz}{1} = \frac{-2xz}{-2y\phi(z)} = \frac{xy - zy^3}{-y^2\,\phi'(z)}$

or $$yz = \frac{xz}{y\phi(z)} = \frac{x - zy^2}{-y\,\phi'(z)} \qquad \text{...(3)}$$

From the first two members, we have $x = y^2\phi(z)$, which is the same as (1).

Thus, we don't get any information about $\phi(z)$.

Taking $$yz = \frac{x - zy^2}{-y\phi'(z)},$$

we have $$\phi'(z) = \frac{x - zy^2}{-y^2z} = -\frac{x}{y^2}\cdot\frac{1}{z} + 1 = -\frac{1}{z}\phi(z) + 1 \qquad \text{[Using (1)]}$$

i.e., $\dfrac{d\phi}{dz} + \dfrac{1}{z}\phi = 1$ which is a linear equation.

$$\text{I.F.} = e^{\int \frac{1}{z} dz} = e^{\log z} = z$$

$\therefore$ $$\phi(z) = \int z dz + c = \frac{z^2}{2} + c \quad \text{or} \quad \phi = \frac{z}{2} + \frac{c}{z} \quad \text{or} \quad \frac{x}{y^2} = \frac{z}{2} + \frac{c}{z}$$

or $$2xy = y^2(z^2 + 2c) \quad \text{or} \quad 2xz = y^2(z^2 + C), \qquad \text{taking } 2c = C$$

which is the required solution.

Example 4. *Solve $2yzdx + zxdy - xy\,(1 + z)\,dz = 0$.*

Sol. Here $\quad P = 2yz,\ Q = zx,\ R = -xy\,(1 + z)$

$\therefore$ $$P\left(\frac{\partial Q}{\partial z} - \frac{\partial R}{\partial y}\right) + Q\left(\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z}\right) + R\left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right)$$
$$= 2yz\,[x + x + xz] + zx\,[-y - yz - 2y] - xy(1 + z)(2z - z)$$
$$= 4xyz + 2xyz^2 - 3xyz - xyz^2 - xyz - xyz^2 = 0.$$

Thus the condition of integrability is satisfied.

Considering z as constant so that $dz = 0$, the given equation becomes $2yzdx + zxdy = 0$

or $$2\frac{dx}{x} + \frac{dy}{y} = 0$$

Integrating $2 \log x + \log y = \log c \quad$ or $\quad x^2y = c$

Replacing c by $\phi(z)$, we have $x^2y = \phi(z)$ ...(1)

Differentiating (1) w.r.t. x, y, z we get $2xydx + x^2dy - \phi'(z)dz = 0$ ...(2)

Comparing (2) with the given equation, we get $\dfrac{2yz}{2xy} = \dfrac{zx}{x^2} = \dfrac{xy(1+z)}{\phi'(z)}$

or $$\frac{z}{x} = \frac{xy(1+z)}{\phi'(z)} \quad \text{or} \quad \phi'(z) = \frac{x^2y(1+z)}{z}$$

or $$\phi'(z) = \phi(z)\left(\frac{1}{z} + 1\right) \qquad \text{[Using (1)]}$$

or $$\frac{\phi'(z)}{\phi(z)} = \frac{1}{z} + 1$$

Integrating, $\quad \log \phi(z) = \log z + z + \log c = \log cze^z$

or $$\phi(z) = cze^z$$

or $$x^2y = cze^z$$

Method III. Method of homogeneous equations. In case P, Q, R are homogeneous functions of x, y, z, then one variable, say z, may be separated from the other two by the substitution $x = uz$, $y = vz$ as illustrated by the following examples.

Example 5. *Solve $(y^2 + yz)\,dx + (z^2 + zx)\,dy + (y^2 - xy)\,dz = 0$.*

Sol. Here $\quad P = y^2 + yz,\ Q = z^2 + zx,\ R = y^2 - xy$

$\therefore$ $$P\left(\frac{\partial Q}{\partial z} - \frac{\partial R}{\partial y}\right) + Q\left(\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z}\right) + R\left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right)$$
$$= (y^2 + yz)\,(2z + x - 2y + x) + (z^2 + zx)(-y - y) + (y^2 - xy)(2y + z - z)$$
$$= 2(y^2 + yz)(x - y + z) - 2y(z^2 + zx) + 2y(y^2 - xy)$$
$$= 2(xy^2 - y^3 + y^2z + xyz - y^2z + yz^2 - yz^2 - xyz + y^3 - xy^2) = 0$$

Thus the condition of integrability is satisfied.

Since, P, Q, R are homogeneous functions of x, y, z, we substitute $x = uz$ and $y = vz$ so that $dx = zdu + udz$ and $dy = zdv + vdz$

The given equation becomes

$$z^2(v^2 + v)(zdu + udz) + z^2(1 + u)(zdv + vdz) + z^2(v^2 - uv)dz = 0$$

or $$z[(v^2 + v)du + (1 + u)dv] + [u(v^2 + v) + v(1 + u) + v^2 - uv]dz = 0$$

or $$z[v(v + 1)du + (1 + u)\,dv] + [uv^2 + uv + v + v^2]dz = 0$$

or $$z[v(v + 1)du + (1 + u)dv] + (v^2 + v)(u + 1)dz = 0$$

or $$\frac{du}{u+1} + \frac{dv}{v(v+1)} + \frac{dz}{z} = 0 \quad \text{or} \quad \frac{du}{u+1} + \left(\frac{1}{v} - \frac{1}{v+1}\right)dv + \frac{dz}{z} = 0$$

Integrating, we have

$$\log (u + 1) + \log v - \log (v + 1) + \log z = \log c$$

or $$\frac{(u+1)vz}{v+1} = c$$

or $$\left(\frac{x}{z} + 1\right)y = c\left(\frac{y}{z} + 1\right)$$

or $$(x + z)y = c(y + z)$$

TEST YOUR KNOWLEDGE

Solve the following equations :

1. $x\,dx + z\,dy + (y + 2z)\,dz = 0$

2. $yz\,dx + zx\,dy + xy\,dz = 0$

3. $(y + z)\,dx + (z + x)\,dy + (x + y)\,dz = 0$

4. $(yz + xyz)\,dx + (zx + xyz)\,dy + (xy + xyz)\,dz = 0$

5. $yz \log z\,dx - zx \log z\,dy + xy\,dz = 0$

6. $2xyzdx - z\,(x^2 + z)dy - y(x^2 + y)dz = 0$

7. $3x^2\,dx + 3y^2\,dy - (x^3 + y^3 + e^{2z})\,dz = 0$

8. $(2x^2 + 2xy + 2xz^2 + 1)\,dx + dy + 2z\,dz = 0$

9. $(yz + z^2)\,dx - xz\,dy + xy\,dz = 0$

10. $yz\,(y + z)\,dx + zx\,(x + z)\,dy + xy\,(x + y)\,dz = 0$

11. $2\,(y + z)\,dx - (x + z)\,dy + (2y - x + z)\,dz = 0$

12. $z\,(z - y)\,dx + z\,(z + x)\,dy + x\,(x + y)\,dz = 0.$

Answers

1. $\frac{x^2}{2} + yz + z^2 = c$ **2.** $xyz = c$ **3.** $xy + yz + zx = c$

4. $\log xyz + (x + y + z) = c$ **5.** $x \log z = cy$ **6.** $x^2 + y + z = cyz$

7. $x^2 + y^2 = e^{2z} + ce^z$ **8.** $(x + y + z^2)\,e^{x^2} = c$ **9.** $xz = c\,(y + z)$ **10.** $xyz = c\,(x + y + z)$

11. $(x + z)^2 = c\,(y + z)$ **12.** $(x + y)\,z = c\,(x + z).$

13.14. SOLUTION OF SIMULTANEOUS EQUATIONS OF THE FORM

$$\frac{\mathbf{dx}}{\mathbf{P}} = \frac{\mathbf{dy}}{\mathbf{Q}} = \frac{\mathbf{dz}}{\mathbf{R}} \text{ where P, Q, R are functions of x, y, z.}$$

(*a*) Method of Grouping

If one variable, say z, is absent or can be cancelled in the equation $\frac{dx}{P} = \frac{dy}{Q}$, leaving the equation in x and y only, integrate it getting $f(x, y) = c_1$

Again, if one variable, say x, is absent or can be cancelled using (1) in the equation $\frac{dy}{Q} = \frac{dz}{R}$, leaving the equation in y and z only, integrate it getting $\phi(y, z) = c_2$.

The two solutions $f(x, y) = c_1$ and $\phi(y, z) = c_2$ give the complete solution of the given equation.

ILLUSTRATIVE EXAMPLES

Example 1. *Solve* $\frac{dx}{xy} = \frac{dy}{y^2} = \frac{dz}{xyz - 2x^2}$.

Sol. Taking the first two members, we have $\frac{dx}{xy} = \frac{dy}{y^2}$ or $\frac{dx}{x} = \frac{dy}{y}$

Integrating, $\log x = \log y + \log c_1$

$$\therefore \quad x = c_1 y \qquad \text{...(1)}$$

Again, taking the last two members, we have $\frac{dy}{y^2} = \frac{dz}{xyz - 2x^2}$

or $$\frac{dy}{y^2} = \frac{dz}{c_1 y^2 z - 2c_1^2 y^2} \qquad \text{[Using (1)]}$$

or $$dy = \frac{dz}{c_1 z - 2c_1^2}$$

or $$c_1 dy = \frac{dz}{z - 2c_1}$$

Integrating, $c_1 y = \log (z - 2c_1) + c_2$

or $$x = \log \left(z - \frac{2x}{y} \right) + c_2 \qquad [\because \; x = c_1 y]$$

or $$x = \log (yz - 2x) - \log y + c_2 \qquad \text{...(2)}$$

Hence (1) and (2) form the complete solution of the given equation.

(*b*) **Method of Multipliers**

We know that $$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} = \frac{l dx + m dy + n dz}{l P + m Q + n R}$$

where l, m, n may be functions of x, y, z or constants.

If the multipliers l, m, n are such that $l P + m Q + n R = 0$

then $$l dx + m dy + n dz = 0$$

which on integration gives $\phi(x, y, z) = c_1$ ...(1)

If it is possible to find another set of multipliers, say l', m', n' such that

$$l' P + m' Q + n' R = 0 \quad \text{then} \quad l' dx + m' dy + n' dz = 0$$

which on integration gives $\Psi(x, y, z) = c_2$ ...(2)

The two solutions (1) and (2) give the complete solution of the given equation.

Note. Sometimes a combination of the two methods is found useful in solving the equation.

Example 2. *Solve* $\dfrac{dx}{x(y^2 - z^2)} = \dfrac{dy}{y(z^2 - x^2)} = \dfrac{dz}{z(x^2 - y^2)}$.

Sol. Using multipliers x, y, z

$$\text{Each fraction} = \frac{xdx + ydy + zdz}{x^2(y^2 - z^2) + y^2(z^2 - x^2) + z^2(x^2 - y^2)} = \frac{xdx + ydy + zdz}{0}$$

$\therefore \quad xdx + ydy + zdz = 0$

Integrating, $x^2 + y^2 + z^2 = c_1$ ...(1)

Using multipliers $\dfrac{1}{x}, \dfrac{1}{y}, \dfrac{1}{z}$

$$\text{Each fraction} = \frac{\frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz}{(y^2 - z^2) + (z^2 - x^2) + (x^2 - y^2)} = \frac{\frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz}{0}$$

$\therefore \quad \dfrac{1}{x}dx + \dfrac{1}{y}dy + \dfrac{1}{z}dz = 0$

Integrating, $\log x + \log y + \log z = \log c_2$ or $xyz = c_2$ ...(2)

Hence the complete solution of the given equation is $x^2 + y^2 + z^2 = c_1$, $xyz = c_2$.

Example 3. *Solve* $\dfrac{dx}{x^2 - y^2 - z^2} = \dfrac{dy}{2xy} = \dfrac{dz}{2xz}$.

Sol. From the last two members, we have $\dfrac{dy}{2xy} = \dfrac{dz}{2xz}$ or $\dfrac{dy}{y} = \dfrac{dz}{z}$.

Integrating, $\log y = \log z + \log c_1$ or $y = c_1 z$ ...(1)

Using multipliers x, y, z, we have

$$\text{Each fraction} = \frac{dz}{2xz} = \frac{xdx + ydy + zdz}{x(x^2 - y^2 - z^2) + 2xy^2 + 2xz^2}$$

or $$\frac{dz}{2xz} = \frac{xdx + ydy + zdz}{x(x^2 + y^2 + z^2)}$$

or $$\frac{dz}{z} = \frac{2xdx + 2ydy + 2zdz}{x^2 + y^2 + z^2}$$

Integrating, $\log z + \log c_2 = \log (x^2 + y^2 + z^2)$

or $x^2 + y^2 + z^2 = c_2 z$ ...(2)

Hence the complete solution of the given equation is

$y = c_1 z$, $x^2 + y^2 + z^2 = c_2 z$.

TEST YOUR KNOWLEDGE

Solve :

1. $\dfrac{dx}{x} = \dfrac{dy}{y} = \dfrac{dz}{z}$

2. $\dfrac{dx}{yz} = \dfrac{dy}{zx} = \dfrac{dz}{xy}$

3. $\dfrac{dx}{z} = \dfrac{dy}{-z} = \dfrac{dz}{z^2 + (x + y)^2}$

4. $\dfrac{dx}{y^2} = \dfrac{dy}{x^2} = \dfrac{dz}{x^2 y^2 z^2}$

5. $\dfrac{dx}{x(y-z)} = \dfrac{dy}{y(z-x)} = \dfrac{dz}{z(x-y)}$

6. $\dfrac{dx}{mz-ny} = \dfrac{dy}{nx-lz} = \dfrac{dz}{ly-mx}$

7. $\dfrac{dx}{x(y^2-z^2)} = \dfrac{dy}{-y(z^2+x^2)} = \dfrac{dz}{z(x^2+y^2)}$

8. $\dfrac{dx}{x^2-yz} = \dfrac{dy}{y^2-zx} = \dfrac{dz}{z^2-xy}$

9. $\dfrac{dx}{x(2y^4-z^4)} = \dfrac{dy}{y(z^4-2x^4)} = \dfrac{dz}{z(x^4-y^4)}$

10. $xdx + ydy + zdz = 0,\ yzdx + zxdy + xydz = 0.$

Answers

1. $x = c_1 y,\ y = c_2 z$

2. $x^2 - y^2 = c_1,\ x^2 - z^2 = c_2$

3. $x + y = c_1,\ \log[(x+y)^2 + z^2] - 2x = c_2$

4. $x^3 - y^3 = c_1,\ x^3 + \dfrac{3}{z} = c_2$

5. $x + y + z = c_1,\ xyz = c_2$

6. $lx + my + nz = c_1,\ x^2 + y^2 + z^2 = c_2$

7. $yz = c_1 x,\ x^2 + y^2 + z^2 = c_2$

8. $\dfrac{x-y}{y-z} = c_1,\ \dfrac{y-z}{z-x} = c_2$

9. $xyz^2 = c_1,\ x^4 + y^4 + z^4 = c_2$

10. $xyz = c_1,\ x^2 + y^2 + z^2 = c_2.$